

TSERING STOB DAN



SEABUCKTHORN

The Superfruit

Unlocking the secrets of a little-
known wonder plant



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*In honour of Prof. Brahma Singh, who taught me
the value of indigenous crops and the wonders of
seabuckthorn*

Preface

"What is Seabuckthorn?"; "Does it only grow in Ladakh?"; "What is so special about seabuckthorn?"; "Can seabuckthorn oil cure cancer?"; "Why is seabuckthorn oil so costly?"; "Which is better - seed oil or pulp oil?"; "Can we grow seabuckthorn without irrigation?" These questions often arise when discussing seabuckthorn, a unique wild plant native to Ladakh and cold regions. Its unique properties, from its ability to grow in harsh conditions to its rich nutritional content, have led me to compile the facts about this remarkable plant into a book.

My personal journey with seabuckthorn began when I joined the Ladakh-based Defence Institute of High Altitude Research (DIHAR). I was astounded to discover the extensive research being conducted by DRDO scientists on this plant, which I had previously only known as a thorny wild bush surrounding our barley fields. This discovery sparked my interest and led to my first article, 'Seabuckthorn: An Emerging Storehouse for Researchers in India', which was published in 2008 on the cover page of Current Science, a prestigious science journal from India. Since then, I have had the privilege of delving deeper into the wonders of this plant and preparing numerous reports.

Why does seabuckthorn remain nature's best-kept secret? Why have only a few people heard about the plant? Based on scientific reports and our research findings, I strongly feel that people, especially the health-conscious ones, should benefit from the thorny bush. This book attempts to showcase why we should regularly use seabuckthorn-based products in everyday life.

Seabuckthorn, a plant of immense medicinal, environmental, and commercial value, holds a transformative potential that is yet to be fully realized. Its berries, among the most nutritious fruits, contain a treasure trove of health benefits. The oil, a prized component, is a rare plant source of omega-3, omega-6, omega-7, and omega-9. What sets seabuckthorn seed oil apart is its unique one-to-one ratio of omega-3 and omega-6 fatty acids. The shrub's medicinal prowess is well-documented, with clinical studies demonstrating the effectiveness of seabuckthorn-based formulations in combating hypertension, obesity, and cardiovascular risk factors. Cell line studies further reveal the plant's potential in cancer treatment and anti-aging. The power of seabuckthorn is truly awe-inspiring- a potential that is waiting to be harnessed and could revolutionize our approach to health and sustainability.

Although not an academic book, it is written with meticulous detail to cater to the health-conscious

reader. Every claim is supported by rigorous scientific studies, providing a solid foundation for the information presented. This approach allows readers to dive deeper into the research if desired, fostering a sense of confidence in the information they are receiving.

I extend my heartfelt gratitude to Ms. Anuja Dasgupta, Dr. Naveen Bisht, and Mr. Phuntsog Tsering for their invaluable contributions to editing this book. Their expertise and dedication have been instrumental in shaping this work. I am also deeply thankful to Dr. Girish Korekar, Mr. Thinles Singhey Rinchen, and Mr. Deldan Namgail for generously allowing the use of their photographs, which have greatly enhanced the visual appeal of this work. Their contributions have been invaluable, and I am truly grateful for their support.

A major part of this book is based on research at the DIHAR over three decades. I thank the Director and colleagues at DIHAR for their constant, unwavering support. I also thank my Ph.D. scholars, who generated new knowledge on topics related to seabuckthorn. Thank you, Geetika and her team. I appreciate your mentoring and showing me how to write a book step-by-step.

This book would not have been possible without the love and support of my family- Dr Dekyi, Chosdan, and Nawang!

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Chapter-1

Seabuckthorn: The Wonder Plant



A berry-laden female seabuckthorn plant

Seabuckthorn is a fascinating thorny shrub that bears nutritious red, orange, or yellow fruits about the size of a pea. The fruits are soft, delicate, juicy, often called berries. The plant typically grows 2-4 m in height and has silvery leaves. It has immense medicinal, environmental, and commercial value. It is naturally found in Asia and Europe's harsh, cold regions. The

shrub grows wild in the high mountainous region of Ladakh in India.

Why the name?

The name 'Seabuckthorn' comes from its habit of growing near the 'sea' and possessing many 'thorns' that is reminiscent of some 'buckthorn' species (of the genus *Rhamnus*). The scientific name is *Hippophae*, which comes from the Latin words 'Hippo' meaning horse, and 'faos' meaning shine. Therefore, the generic name *Hippophae* is classical Latin for 'shining horse'. It is believed that Alexander the Great's army noticed that feeding sick and injured horses with seabuckthorn leaves and berries improved their health and made their coat shiny (Small et al., 2002; Doukas, 2023).

Seabuckthorn is known in different languages as *Shaji* in Chinese, *Duindoorn* in Dutch, *Sanddorn* in German, *Olivello Spinoso* in Italian, *Oblepicha* in Russian, *Tyrni* in Finnish, *Espino de Mar*, *Falso Espino*, *Espino Amarillo* in Spanish, *Havtorn* in Swedish, *Yalanci igde*, *Karga diken* in Turkish, *sTar-Bu* in Tibetan, *Ghangaru* in Nepali, *Torobo* in Bhutia. The fruits have a distinctive sour taste and a unique aroma reminiscent of pineapple. Indeed, in Belarus, the fruit juice is known as Russian pineapple.

In India, it is known as *tSer-Mang*, *tSer-Ta-Lu-Lu*, *Shib-Shu-Lu-Lu* in Ladakh; *Amesh*, *Chuk*, *Amil*, and

Tarwar in Uttarakhand; and *Chharma* in Himachal Pradesh. The plant is identified as the Ayurvedic medicinal plant *Amlich* or *Badriphal* (Sharma & Chuneekar, 1998)

Amazing Facts

There are seven species of *Hippophae*, all of which are called seabuckthorn. *H. rhamnoides* is the most widely distributed species, while the remaining species have limited distribution. Seabuckthorn is widely distributed across various geographical areas, with altitudes ranging from a few meter up to 5,200 m (Rongsen, 1992).

Seabuckthorn has a long history of application in Tibetan, Mongolian, and Chinese systems of medicine. Current research is now beginning to understand and support the traditional uses. Seabuckthorn is extremely useful in land stabilization and reclamation. However, its commercial value lies in its nutritional, cosmetic, and medicinal qualities (Li & Beveridge, 2007).

Seabuckthorn berry is one of the most nutritious fruits and has immense medicinal properties. It has the commercial status of a novel 'Super-fruit' (Kalia et al., 2011). Berries contain more than 100 different nutrients and bioactive substances. The berry is too acidic to consume directly. Its pH range is 2.7-3.1 (Seglina et al., 2006).

Vitamin C is a nutrient of significant importance in seabuckthorn because of its presence in large quantities. The vitamin C contents are higher than those of aonla and orange. There is enough vitamin C in seabuckthorn berries around the world to meet the dietary requirements of the entire human population (Small et al., 2002).

The most valuable component of the berries is the oil. Two types of oil can be extracted: pulp oil and seed oil. Seed oil is commonly used in cosmetic preparations, such as facial creams. Seabuckthorn berry products are popular functional foods in the USA, Canada, and Europe, with over 200 products developed worldwide from the seabuckthorn (Kallio et al., 2002).

The plant is known for its hardiness and can withstand extreme temperatures ranging from -43°C to 40°C (Rongsen, 1992). Seabuckthorn is not a leguminous crop. However, the shrub develops an extensive root system that can fix atmospheric nitrogen, improving soil fertility.

There are two types of plants: male and female. Only the female plants produce fruits. Neither the male nor the female flowers contain nectar. The flowers are small and not very colorful or fragrant, so they don't attract pollinating insects. As a result, female flowers rely entirely on wind to carry pollen from the male flowers for pollination. Commercial production of berries relies heavily on female plants, while only 10%

of male plants are required in the field to produce ample fertile pollen.

The ratio of male and female plants is 1:1 under optimal growth conditions. However, male plants dominate in harsh and resource-limited environmental conditions (Li et al., 2007). Stressful environments have a more detrimental impact on females than males (Dolkar et al., 2019). The leaves from male plants exhibit higher antioxidants than those from females. October is the best time to harvest the leaves for their higher health-promoting compound content (Dolkar et al., 2017a).



A male seabuckthorn plant

The abscission process remains blocked in seabuckthorn fruit, meaning the ripe berries can stay on the shrub for up to three years if not disturbed.

However, applying a high ethylene concentration (0.5 ml per l) can remove the blocked abscission system (Demenki, 2003).

Several wildlife species depend on the seabuckthorn berries for food and shelter. In Ladakh, the berry is often called *Chepay-zanma*, i.e., 'food of birds'. The berries remain on the shrub throughout the winter and serve as one of the primary sources of food for birds during winter (Stobdan et al., 2013).



Seabuckthorn berries remain on the plant in winter

The shrub is suitable for many demanding environments where most plants wouldn't survive. Several parts of the world have witnessed successful cases of environmental restoration by planting seabuckthorn.

Seabuckthorn has many diverse uses. All plant parts, including berries, leaves, twigs, roots, and bark, have

traditionally been used for medicine, nutritional supplements, firewood, and fences.

Seabuckthorn's total acreage is about 2.3 million hectares worldwide (wild and cultivated cover). It is primarily found in China, Mongolia, Romania, India, Russia, and Pakistan (Nybom et al., 2023). The domestication of seabuckthorn began in Siberia in the 1930s (Kalinina & Panteleyeva, 1987).

In India, seabuckthorn received much attention after the Defence Institute of High Altitude Research (DIHAR) developed and commercialized the production of ready-to-serve beverages from the highly acidic fruit. The drink was introduced in the Indian market under the 'Leh Berry' brand. Due to the brand's popularity, seabuckthorn is often called the Leh Berry plant. Once considered a thorny menace, the shrub is now viewed as a means for sustainable development in mountain regions.

Historical Account

Seabuckthorn is mentioned in ancient Greek writings by scholars such as Theophrastus and Dioscorides (Rongsen, 1992). It is believed to be the favored food of Pegasus, the winged flying horse of Greek mythology (Small et al., 2002).

The medicinal benefits of seabuckthorn have been known since the 8th century, as documented in the

Tibetan medicinal classic *rGyud Bzi* (Rongsen, 1992). Seabuckthorn oil is approved for clinical use in China, where it was formally listed in the Pharmacopoeia in 1977 (Xu, 1994). Seabuckthorn was used to treat the victims of the 1986 Chernobyl nuclear disaster in Russia (Small et al., 2002).

Seabuckthorn is credited as one factor contributing to the Mongolian Genghis Khan's conquest of a large part of the world during the 12th and 13th centuries (Letchamo et al., 2007). Seabuckthorn drink was the official beverage for Chinese athletes at the Seoul Olympic Games in 1988 (Small et al., 2002).

Commercial cultivation of seabuckthorn started in Russia during the 1920s with the development of cultivars for large commercial farms (Letchamo et al., 2007). The seabuckthorn industry has thrived in Russia since 1940 (Li, 2002). The development of superior seabuckthorn cultivars was a closely guarded secret under the Russian space program during the space race of the 1960s and 1970s (Letchamo et al., 2007). Russian cosmonauts were supplied with seabuckthorn beverages to enhance their health and resistance to stress. It was the first fruit juice in space (Small et al., 2002). Russian cosmonauts used seabuckthorn oil to protect against harmful radiation.

The International Seabuckthorn Association (ISA) was founded in 2001 to promote and develop seabuckthorn.

General description of the plant

Growth Habits and Height: Seabuckthorn (*H. rhamnoides*) is a bush or small tree growing up to 1-4 m tall. It can grow up to 15-18 m in forest areas, especially along river banks (Rongsen, 1992). However, *H. tibetana* grows only 7-60 cm in height, while *H. salicifolia* is tall and usually grows 4-5 m in height.

Lifespan: The natural lifespan of the seabuckthorn plant is 60-70 years. Fruit-bearing trees over 300 years old are also found in China (Rongsen, 1992).

Flowering: Seabuckthorn is predominantly a dioecious species, i.e., those that have separate male and female plants. Flowering occurs when the plants are entirely leafless between the first week of April and the first week of May.



Golden-dark brown male flower

Seabuckthorn: The Superfruit

The flowers lack nectar and fragrance. The female flowers depend almost entirely on the wind for pollination. Pollen can travel up to 15 m (Mangla & Tandon, 2014).

Fruit: The fruit of seabuckthorn is drupe-like, consisting of a single seed surrounded by a fleshy hypanthium. The fruits are small, soft, delicate, and juicy. The fruits are round to slightly elongated red, orange, or yellow, and remain on the shrub throughout the winter. The fruit ripens in late August to mid-September.



Current and previous year's berries on the shrub

Seed: Seabuckthorn seeds are hard, dark brown, 3-5 mm in length, 1.5-2.5 mm thick. Fresh seabuckthorn seed oil crystallizes at -59°C , a temperature much

lower than those of vegetable oils such as canola (-43°C) and sunflower (-45°C) (Beveridge et al., 1999).

Leaves: Seabuckthorn has silvery deciduous leaves. The leaves are alternate, 2-4 mm wide, with silver to gray color on the upper side, and under the surface is covered with peltate hairs (Rousi, 1971).



Difference in leaf sizes of *H. salicifolia*, *H. rhamnoides*, and *H. tibetana*

Root: The shrub develops an extensive root system that can fix atmospheric nitrogen. In a 10-12-year-old shrub, the horizontal root extends up to 5.4 m and vertically up to 1.3 m deep, thus making it an ideal plant for soil binding (Dwivedi et al., 2006).

Chapter-2

Global and Regional Distribution



Berry-laden seabuckthorn bush in winter

Seabuckthorn is native to Asia and Europe. Its distribution is extensive throughout various geographical areas of the world. It is naturally distributed over Asia and Europe's arid, semi-arid, and high mountainous ecosystems. *H. rhamnoides* is the most commonly found species, followed by *H. salicifolia* and *H. tibetana*.

Seabuckthorn grows in various landscapes, including mountains, hills, valleys, riverbeds, and coastal regions. It can grow in small or isolated stands or mixed with other shrub or tree species. The shrub grows as a wild plant in the high mountainous Ladakh region in India. Natural seabuckthorn is found on European riverbanks and coastal dunes, including the Baltic coasts of Germany, Finland, Poland, the Gulf of Bothnia in Sweden, and the UK coastal regions (Rousi, 1971).

Environmental Requirements

Seabuckthorn is widely distributed and can survive in areas with an annual mean temperature ranging from 2.7°C to 10.7°C.



Hippophae tibetana with its distinct star-shaped mark on the berries

However, young seedlings are often killed by high temperatures above 30°C. *H. tibetana* is scattered throughout the cold regions where the annual mean temperature ranges from -4.7°C to 1.1°C (Rongsen, 1992).

Most natural populations grow in areas receiving 400–600 mm of annual precipitation. Soil moisture is critical in spring, during which flowering and young fruit development occur. Therefore, areas receiving 600–700 mm of annual precipitation are preferred for commercial cultivation. Assured irrigation is required if the area receives less than 400 mm of annual precipitation (Rongsen, 1992). Mountain regions between 1,500–3,000 m above sea level are the most suitable sites for seabuckthorn plantations.

Distribution

Seabuckthorn is distributed in 52 countries worldwide, covering a total area of 2.33 million hectares. China accounts for approximately 2.1 million hectares, with the remaining area distributed across other countries. The other major seabuckthorn producing countries are Mongolia (20,000 hectares), Romania (15,000 hectares), India (14,500 hectares), Russia (60,00 hectares), and Pakistan (5,700 hectares) (Nybom et al., 2023). The shrub is also found in Afghanistan, Azerbaijan, Armenia, Belarus, Britain, Bhutan, Finland, France, Germany, Hungary, Italy, Kazakhstan, Nepal, Norway, Poland, Sweden,

Switzerland, Ukraine and Uzbekistan. The plant has also been introduced to North America.

China: Seabuckthorn is well adapted to different geographical and climatic conditions in China and can grow from 420-5,200 m above sea level (Singh, 2022). The shrub is primarily distributed in northern regions, including Liaoning, Inner Mongolia, Hebei, Shanxi, Shaanxi, Ningxia, Qinghai, Gansu, and Xinjiang. Additionally, it is found in high-altitude areas such as Tibet, Sichuan, and Yunnan (Mei et al., 2023). The area under seabuckthorn is approximately 2.1 million hectares, most of which have been planted (1.35 million hectares) (Nybom et al., 2023). There are six species and eight subspecies of seabuckthorn in the country, earning it the title "Kingdom of Sea Buckthorn". *H. rhamnoides* is the most dominant species (Mei et al., 2023).

Mongolia: *H. rhamnoides* is found in Mongolia. The species covers about 20,000 hectares of land. It is mostly found in Selenge, Bulgan, Zavkhan, Gobi-Altai, Khovd, and Uvs provinces (Nybom et al., 2023; Oyungerel et al., 2014).

India: The plant is widely distributed in Ladakh and pockets of Himachal Pradesh, Uttarakhand, Sikkim, and Arunachal Pradesh. The approximate area covered by seabuckthorn is 14,500 hectares. Ladakh remains the major site for natural seabuckthorn resources,

with over 64% of the total area under seabuckthorn in the country.

Ladakh: According to satellite data, there are approximately 7,184 hectares of pure seabuckthorn in the area, while the area under mixed seabuckthorn is estimated to be 2,083 hectares (Chauhan et al., 2006). Although two species viz. *H. rhamnoides* and *H. tibetana* are found in this region, most of the area is under *H. rhamnoides*. Plants with characteristics of *H. salicifolia* in the region are found in the Suru Valley in the region (Stobdan et al., 2017). The plant grows in a wide range of soil, ranging from highly acidic soil to soil with high salt deposits (Singh, 2004). The shrub is primarily distributed in the Nubra, Indus, and Suru Valleys. It naturally grows along the Nubra, Shyok, and Indus Rivers, as well as in the Changthang region, at an altitude of 4,700 m (Stobdan et al., 2009).



Seabuckthorn shrubs in natural habitat in Ladakh

Uttarakhand: *H. salicifolia* is commonly found between 1,700–3,000 m elevations, particularly along rivers and tributaries originating from glaciers. It grows in Uttarkashi, Rudraprayag, Chamoli and Pithoragarh districts (Yadav et al., 2009). *H. tibetana* is found at an altitude of 3,300–4,500 m in Pithoragarh and Chamoli districts (Singh, 2022). The total area covered by seabuckthorn is approximately 3,750 hectares (Yadav et al., 2009).

Himachal Pradesh: Wild populations of three species of seabuckthorn, namely *H. rhamnoides*, *H. salicifolia*, and *H. tibetana*, have been documented in the Lahaul-Spiti and Kinnaur districts. It is found scattered in villages like Sissu, Khorpani, Raling, Khongsar, Gondhla, and Dalang in the Chandra Valley (2,958–3,110 m) in Lahaul. A stand of *H. rhamnoides* is found along the Bhaga River from Gemur to Darcha. *H. salicifolia* is found in Kardang and Tinu villages. In the Spiti Valley, seabuckthorn grows from Sumdo village (3,100 m) to upper Spiti (3,900 m). *H. tibetana* grows at higher altitudes such as the Losar (4,100 m), Takcha (4,500 m), Kibbar (4,100 m), and Chicham (4,200 m) regions (Singh 2022). Based on preliminary assessment, the total area covered by seabuckthorn is estimated to be around 1,000 hectares (Stobdan et al., 2011a).

Sikkim: Seabuckthorn grows widely on slopes and riverbanks around the Lachen, Zema I, Zema II, Zema III, and Lachung regions (Basistha, 2009). It covers an estimated forest area of 300–500 hectares. The

dominant species is *H. salicifolia*, which is distributed between 2,280–3,100 m (Singh, 2022). Small patches of *H. rhamnoides* are found in the upper zone, ranging from 2,800–3,300 m (Basistha, 2009).

Arunachal Pradesh: Seabuckthorn grows in small patches along the banks of the Naymjang Chu River in the Zimathang area, which is about 100 km from Tawang. There is a stretch of about 5 km of seabuckthorn along the bank at 2,438 m altitude in the area (Vairale et al., 2020).

Conservation Efforts

Despite being a widespread species, seabuckthorn needs conservation. Its distribution is highly fragmented and tends to occur in isolated patches. For example, the unusually hairy *H. gyantsensis* occurs only in a restricted portion of Tibet. The nutritional and medicinal value of the shrub has led to uncontrolled exploitation and destruction of seabuckthorn resources in some parts of its natural distribution. Thus, the protection and preservation of the shrub have become urgent. In Hungary, the plant is protected as an endangered species (Small et al., 2002).

Man-made forest fires are a major cause of the destruction of the natural seabuckthorn habitat. Fire is caused when a source of fire, such as a naked flame, cigarette, electric spark, or any source of ignition,

comes into contact with inflammable material. Therefore, a forest fire line needs to be established to prevent fire from breaking into the forest from one compartment to another.

Seabuckthorn is a highly versatile and valuable shrub growing in 52 countries. In India the plant is widely distributed in Ladakh and pockets of Himachal Pradesh, Uttarakhand, Sikkim, and Arunachal Pradesh. Ladakh remains the major site for natural seabuckthorn resources, with over 64% of the country's total area under the crop. The benefits of seabuckthorn have been recognized worldwide, which is why it has been introduced to North America and other countries.

Chapter-3

Nutritional Value



Seabuckthorn berries are among the most nutritious fruits, with numerous health benefits. The fruit juice is packed with essential compounds such as vitamins, organic acids, amino acids, phytosterols, flavonoids, and mineral elements. Seabuckthorn contains 4-20 times more phytosterols than soybean oil, which helps lower plasma cholesterol. The oils extracted from seabuckthorn are highly valuable, with the seed oil having a one-to-one ratio of omega-3 and omega-6 fatty acids, making it highly beneficial for overall

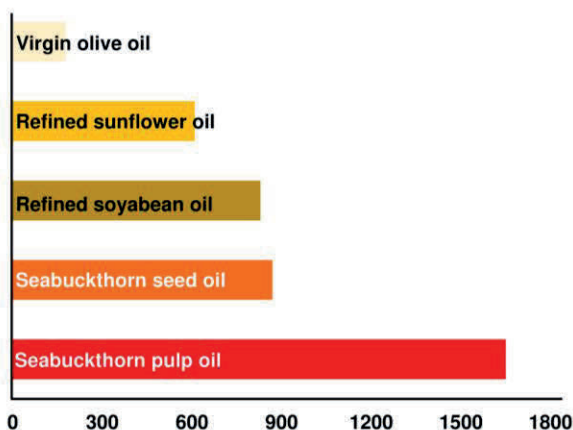
health. Palmitoleic acid is an omega-7 monounsaturated fatty acid uncommon within the plant kingdom. However, seabuckthorn is one of the few plants with this fatty acid in the soft part of the berries. This fatty acid has a strong relation with human health. The seed and pulp oil are also rich sources of omega-9 fatty acid (oleic acid) (Marsiñach & Cuenca, 2019).

Vitamins

Seabuckthorn is known for its extraordinarily high vitamin C (ascorbic acid) level. The content of vitamin C is affected by diverse factors such as origin, harvesting time, geographical location, and processing. The vitamin C concentration in the berries from Ladakh ranges from 56-3,909 mg per 100 g, with a mean value of 998 mg per 100 g of fresh weight (Korekar et al., 2014). Considering that fresh orange juice contains 35-56 mg (Araujo, 1997) and Aonla contains 478.5 mg per 100 ml (Jain & Khurdiya, 2004), the value of seabuckthorn as a source of vitamin C is apparent. The ascorbic acid content in the berries is over 100 times greater than that of fruits such as apples, apricots, bananas, and peaches. The berries have three times higher vitamin C than kiwi, six times higher than citrus, and 80 times higher than tomatoes (Xiao-Hua et al., 2007). Seabuckthorn berries do not contain the ascorbate oxidase, the enzyme responsible for the degradation of ascorbic acid, and therefore, seabuckthorn products and dried berries contain large

amounts of vitamin C (Krejcarova et al., 2015). It is estimated that there is enough vitamin C in the berries of seabuckthorn plants worldwide to meet the dietary requirements of the entire human population (Small et al., 2002).

Seabuckthorn berries are not just a rich source of vitamin C but also contain significant amounts of vitamin E (56-140 mg per kg) (Kallio et al., 2002), surpassing the levels found in tomato paste (50 mg), blanched frozen spinach (42 mg), green bottled olives (38 mg), blackberries (37 mg), raspberries (35 mg), and avocado (31 mg per kg) (Chun et al., 2006). Vitamin E is very effective in preventing and reversing various disease complications. It acts as an antioxidant, reduces inflammation, prevents platelet aggregation, and enhances the immune system.



Significantly higher α -tocopherol (mg per kg) contents in seabuckthorn oil than other oil

Symptoms of vitamin E deficiency include muscle weakness, vision problems, immune system changes, numbness, difficulty walking, and a poor sense of balance. Apart from these symptoms, deficiency can also lead to neuromuscular problems, anemia, and impairment of immune response. Therefore, it is crucial to maintain adequate levels of vitamin E in the body to ensure optimal health (Rizvi et al., 2014).

The berries contain eight different forms of vitamin E, including four tocopherols and four tocotrienols. The most abundant form of vitamin E found in seabuckthorn berries is α -tocopherol (43-116 mg), followed by γ -tocopherol (2.2-15 mg), β -tocopherol (1.3-17 mg), and δ -tocopherol (0.2-3.4 mg per kg of fresh weight). The total tocotrienols content is low (1.5-8.1 mg per kg of fresh weight) compared to the total tocopherols. The vitamin E content of both the pulp (1,640-2,330 mg) and seed oil (872-2,910 mg per kg) is significantly higher than the whole seabuckthorn berries (Kallio et al., 2002). The α -tocopherol in seabuckthorn pulp and seed oil is 1,360-1,940 mg and 444-1294 mg per kg, respectively, and thus significantly higher than the total tocopherol contents in virgin olive oil (177 mg), refined soyabean oil (829 mg), refined sunflower oil (609 mg) and peanut refined oil (226 mg per kg) (Gliszczyńska-Świgło et al., 2007). Even though tocopherols in synthetic forms are available, it is better to consume tocopherols from natural sources. In nature, all α -tocopherol is in the RRR-conformation, but synthetic (all-racemic) α -tocopherol is an

equimolar mixture of the eight possible stereoisomers (RRR, RRS, RSR, RSS, SRR, SSR, SRS, and SSS). Notably, RRR α -tocopherol is more bioavailable and biopotent than all-racemic α -tocopherol. However, the synthetic stereoisomers accumulate in tissues to varying degrees and may elicit different biological effects than RRR α -tocopherol (Ranard & Erdman, 2017; Ranard et al., 2019).

Seabuckthorn pulp is also a rich source of riboflavin (1.45 mg per 100 g), niacin (68.4 mg per 100 g), pantothenic acid (0.85 mcg per 100 g), vitamin B-6 (1.12 mg per 100 g), and vitamin B-2 (5.4 mcg per 100 g), surpassing the levels found in many other fruits. A single fruit type with such a diverse range of vitamins is particularly beneficial in cold desert high-altitude environmental conditions. The challenging terrain and logistical constraints often make it difficult to access fresh fruits and vegetables, especially during winter months. In such situations, seabuckthorn products can serve as a locally available and viable alternative to meet nutritional requirements in high-altitude environments, providing a sense of reassurance and confidence (Stobdan et al., 2010).

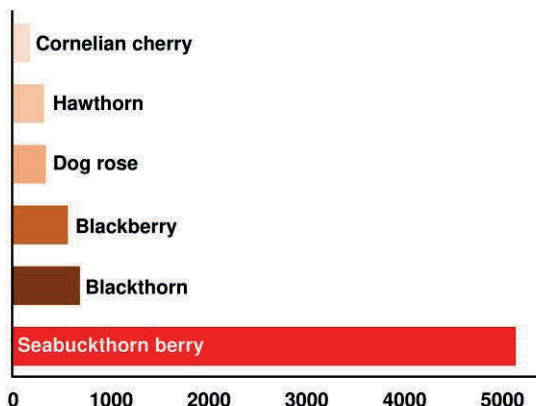
Carotenoids

The color of seabuckthorn berries ranges from yellow to red shades. Their color depends on the carotenoid pigments. Seabuckthorn berries are known as a rich source of carotenoids. The major carotenoids are β -

carotene followed by lycopene and zeaxanthin. The total carotenoids in berries range from 15-185 mg per kg of fresh weight (Andersson et al., 2009). Since the carotenoids are concentrated in the pulp oil, its contents range from 692-3,420 mg per kg of oil (Ranjith et al., 2006). In comparison, red palm oil, which is considered a rich source of carotenoids, contains 600 mg per kg of oil (Nagendran et al., 2000). Up to 18 different carotenoids have been identified in seabuckthorn oil, with the carotenoids possessing provitamin A activity (β -carotene, γ -carotene, β -zeacarotene, β -cryptoxanthin, and sintexanthin) and lutein accounting for 48 and 14% of the total carotenoids, respectively (Li & Beveridge, 2003).

Flavonoids

Flavonoids are phenolic compounds that have a potential role in managing chronic diseases such as diabetes, cardiovascular diseases, and cancer. Their concentration in seabuckthorn berries ranged from 1,680-8,590 mg per kg of fresh weight (Heinaabo & Julkunen-Tiitto, 2011), which is many folds higher than the contents in high-flavonoid plants such as hawthorn (314 mg), cornelian cherry (173 mg), wild European blackberry (558 mg), blackthorn (682 mg), and dog rose (333 mg per kg of fresh weight) (Cosmulescu et al., 2017). Among flavonoids, flavanols are the most abundant subgroup in seabuckthorn.



Significantly higher flavonoid (mg per kg) contents in seabuckthorn berry than other fruits

Phytosterols

Phytosterols have structural similarities to cholesterol, which, on consumption, can lower the plasma cholesterol in humans. Seabuckthorn contains 17 phytosterol compounds (Li et al., 2007). The total quantity of phytosterol in seabuckthorn is remarkably high, potentially surpassing soybean oil by 4-20 times (Li, 2002). The phytosterol content of the oil ranges from 330-550 mg per kg (Sabir et al., 2005). The total sterol content in the fresh soft parts varies within a range of 240-400 mg per kg. In the pulp oil, it ranges from 13.3-28.7 g per kg. The predominant phytosterol in seabuckthorn is sistosterol, constituting 57-76% and 61-83% of seed and soft-tissue sterols, respectively.

Other significant phytosterols include campesterol, stigmastadienol, stigmastanol, and α -amyrin (Yang et al., 2001).

Seabuckthorn seeds, rich in sterols, offer potential health benefits. The total sterol content in the seed varies from 1,200-1,800 mg per kg, of which 57-76% is sitosterol and 13-21% isofucosterol. Stigmastanol, citrostadienol and campesterol constitute 1-5, 2-5 and 2-3% of total seed sterols, respectively. The proportion of other sterols is typically 1-2% each (Yang et al., 2001).

Protein and Amino acids

The protein content of seabuckthorn berries ranged from 46-129 g per kg of dry weight (Ranjith et al., 2006), which is significantly higher than other fruits such as strawberries (18.5 g), raspberries (33-44 g), blackberries (65.6 g per kg of dry weight). The seed contains 38% protein. Protein is one of the important components in seabuckthorn leaves that has value in animal feed and can be used as a source of unconventional protein for human consumption. Leaves of female and male plants are reported to contain an average of 21.1 and 20.6 g protein per 100 g dried leaf, respectively. Leaf protein content peaks during the middle of July and early August and decreases significantly in the middle of August (Li & Wardle, 2003).

Seabuckthorn is rich in various free amino acids; 18 kinds of free amino acids have been detected in the juice, of which eight (threonine, valine, methionine, leucine, lysine, tryptophan, isoleucine, phenyl alanine) are essential for the human body. The most predominant free amino acid in seabuckthorn berries is aspartic acid (426.6 mg) followed by proline (45.2 mg), threonine (36.8 mg), serine (28.1 mg), lysine (27.2 mg), valine (21.8 mg), alanine (21.2 mg), phenyl alanine (20.0 mg), glutamic acid (19.4 mg), isoleucine (17.4 mg), glycine (16.7 mg), histidine (13.7 mg), tyrosine (13.4 mg), arginine (11.3 mg), cysteine (3.3 mg) and methionine (2.3 mg per 100 g) (Zhang et al., 1989). Since the full range of amino acids is present, seabuckthorn juice is a source of high-quality plant-based protein.

Organic Acids

Seabuckthorn berries contain organic acids, mainly malic (3.7 g) and quinic acids (1.5 g per 100 ml juice), which together constitute around 98% of the fruit acids (Yang et al., 2009). Other acids include citric acid, tartaric acid, and succinic acid. Acid content correlates negatively with the sweetness of seabuckthorn berries. High malic acid content is an important contributor to the sour and astringent tastes of seabuckthorn berries and juice (Tiitinen et al., 2005).

Minerals

Seabuckthorn juice contains at least 24 minerals, including calcium, magnesium, phosphorus, iron, manganese, sodium, potassium, and boron. Potassium is the most abundant of all the elements in the berries. Mineral composition of berries from the Ladakh region in the Indian trans-Himalayas revealed high amount of minerals including potassium (647.2 mg), calcium (176.6 mg), iron (30.9 mg), magnesium (22.5 mg), phosphorous (84.2 mg), sodium (414.2 mg), zinc (1.4 mg), copper (0.7 mg), manganese (1.06 mg) and selenium (0.53 mg per l) (Stobdan et al., 2010). The seabuckthorn contents of calcium, iron, sodium, zinc, and manganese are much higher than mango, apricot, banana, orange, and peach. The requirement of minerals is generally high in high altitude conditions because of the increased excretion of electrolytes. Using seabuckthorn as a nutritional supplement can achieve a normal balance of most ions.

Fiber

Dietary fiber is recognized for its health benefits. Soluble and insoluble fibers make up the two basic categories of dietary fibre. Cellulose, hemicellulose, and lignin are not soluble in water, whereas pectins, gums, and mucilages become gummy in water. Seabuckthorn is high in fiber content, and contains 62 g fiber per kg of dry weight (Jaroszewska et al., 2018).

The berries contain 3.9-8.3 g of dietary fiber per kg (Usabalieva et al., 2022).

Fatty acid, Tocopherol, and Carotenoid in oil

Seabuckthorn oil is a valuable product used as a nutraceutical supplement. Oil extracted from seed and pulp differs considerably in fatty acid composition. Oil from juice and pulp is rich in palmitic (16:0) and palmitoleic (16:1) acids, while the oil from the seed contains essential fatty acids, which are linoleic (18:2) and linolenic (18:3) acids. One of the many unique features of the seabuckthorn fruit is the exceptionally high content of tocopherols and tocotrienols. The total content of tocotrienols varied from 1.5-8.1 mg per kg in berries and from 43-188 mg per kg in berry oil (Kallio et al., 2002). The content of these bioactive compounds is among the crucial criteria defining the quality of seabuckthorn. Oil derived from juice contains more vitamin E (216 mg) than seed (64.4-92.7 mg per 100g) (Li, 2002). The tocopherol fraction consists of approximately 50% α -tocopherol, 40% β -tocopherol, and 10% γ -tocopherol (Mironov, 1989). The high content of α -tocopherol has a significant healthful effect, which acts as a natural antioxidant in the human body. The carotenoid content of seabuckthorn oil ranges from 314-2,139 mg per 100 g (Zhang et al., 1989). It is indicated that the carotenoids consist of approximately 20% β -carotene, 30% γ -carotene, 30% lycopene, and 15% oxygen-containing carotenoids. Phytosterols are also constituents of

seabuckthorn oil, which can lower plasma cholesterol when consumed. The major phytosterol is sitosterol (β -sitosterol), followed by 5-avenasterol.

Two main families of polyunsaturated fatty acids (PUFAs) are relevant to human health: the omega-6 and the omega-3 PUFAs. In most diets, omega-6 is present in the highest amount in the form of linoleic acid (LA), while omega-3 is in alpha-linolenic acid (ALA). Animals cannot synthesize LA and ALA, which must be obtained through plant-based sources. Low intake of both LA and ALA can lead to various health problems such as dermatitis, cardiovascular diseases, renal hypertension, type 2 diabetes, impaired brain development, arthritis, depression, and decreased immunity (Djuricic & Calder, 2021). Throughout evolution, humans have consumed a diet in which the ratio of omega-6 and omega-3 essential fatty acids was about 1-2/1. However, with the advent of modern agriculture and agribusiness, the intake of omega-3 fatty acids has decreased while the intake of omega-6 fatty acids has increased. Western diet is known for its significantly higher intake of omega-6 fatty acids than omega-3 fatty acids, with a ratio ranging from 10 to 20 times more. In rural India, the ratio of omega-6 to omega-3 fatty acid intake is around 5-6.1/1, while in modern urban India, it is much higher, at around 38-50/1 ratio. Excessive amounts of omega-6 PUFA and a very high omega-6/omega-3 ratio, as is found in most modern diets, promote the pathogenesis of many chronic diseases, including cardiovascular disease and

cancer. A diet rich in omega-3 PUFA (i.e., a diet with a low ratio of omega-6/omega-3) is associated with a decreased risk of cancer, suppressed inflammation in patients with rheumatoid arthritis, and is beneficial to patients with asthma. For preventing and managing chronic disease, it is desirable to follow a diet with an optimal ratio of omega-6 to omega-3 fatty acids, preferably 1-2/1, similar to those seen in Paleolithic and traditional diets. Furthermore, maintaining a healthy balance of omega-6 and omega-3 fatty acids is crucial for normal development and homeostasis (Simopoulos, 2004). A unique characteristic of the seabuckthorn seed oil is its one-to-one ratio of omega-3 and omega-6 at approximately 34% and 31%, respectively (Bal et al., 2011). Seabuckthorn seed oil is commercially available in capsule form and should be consumed.

Sexual Differences in Polyphenol Contents and Antioxidants

Seabuckthorn leaves are utilized to extract health-promoting compounds and develop products. The total phenolic content is 5.8% higher in male plant leaves compared to female leaves. Similarly, male leaves have higher antioxidant levels than female leaves. Therefore, male leaves are preferred for preparing antioxidant and phenolic-rich products. The content of polyphenols and antioxidants rises from July to October and decreases from November onwards. As a result, October is the best time to

harvest the leaves for higher health-promoting compounds (Dolkar et al., 2017b).

Seabuckthorn is considered one of the best sources of vitamin C and tocopherol. It is also one of the few plant sources that contain omega-3, omega-6, omega-7, and omega-9. What makes seabuckthorn seed oil unique is its one-to-one ratio of omega-3 and omega-6 fatty acids. Also, omega-7 fatty acids are rare in the plant kingdom but are found in the soft part of seabuckthorn berries. The high quantity of phytosterols present in seabuckthorn can help lower human plasma cholesterol. The nutrient-rich content of these berries may be one of the reasons why they are credited with contributing to the conquests of Mongolian Genghis Khan. Although seabuckthorn has been used in natural medicines for many years, it is now gaining popularity as a superfood.

Chapter-4

Medicinal and Therapeutic Uses



Overwintering berries

Seabuckthorn has a fascinating medicinal history that goes way back to the Greeks and Romans. It is used extensively in Tibetan, Mongolian, and Chinese systems of medicine. The shrub's medicinal benefits are well documented in the Tibetan medicinal classic *rGyud Bzi*. Current research is now beginning to understand and support the traditional uses.

Scientific studies have shown that this plant's claimed medicinal properties are supported by clinical and cell line studies. Due to the efficacy demonstrated by

scientific research, the general public is becoming increasingly interested in the medicinal use of seabuckthorn.

Seabuckthorn in Traditional Medicine

Seabuckthorn has a rich history of healing properties and has been used for centuries in traditional medicine. Descriptions of its usage can be found in ancient Tibetan, Chinese, Mongolian, and Ayurvedic texts. The plant's therapeutic values have now been scientifically proven.

The oil extracts from the berries are used to treat liver diseases, inflammation, and absorption disorders in the gastrointestinal system and applied externally for hemorrhage (Bazaron & Chibikova, 1978). The oil is used in the treatment of eczema, lupus erythematosus, chronic wounds that are difficult to heal, inflammatory diseases, erosion of the cervix uteri, the treatment of burns and frozen parts of the body (Chopik et al., 1983). Berry oil also treats skin diseases (Yang et al., 2000) and thrombosis (Cheng et al., 2003).

In Turkey, seabuckthorn has been traditionally used as an antitussive and aphrodisiac (Cakilcioglu et al., 2011). The fruit is used to treat pulmonary, gastrointestinal, cardiac, blood, and metabolic disorders in Tibet, India, and China (Bazaron & Aseeva, 1984; Badareva, 1985). In

Tajikistan, flowers are commonly used as natural skin softeners (Barnaulov et al., 1988).

Extracts of leaves and branches are used to treat colitis and enterocolitis in both humans and animals in Mongolia (Tsybikova et al., 1983). The parts also treat diarrhea (Vereschhagin et al., 1959). The leaves are also used in gastrointestinal and dermatological disorders (Tsybikova et al., 1983) and in the treatment of rheumatoid arthritis (Khalmatov et al., 1984).

Seabuckthorn in the Tibetan System of medicine

Seabuckthorn is commonly used in *Sowa Rigpa*, the Tibetan system of medicine. It is also known as the *Amchi* system of medicine. The medicinal properties of seabuckthorn have been documented since the 8th century in the Tibetan medicinal classic, *rGyud Bzi* (Four Medical Tantras). It is also mentioned in the classical medical text, the *dTrimed Shelgong* and the *Shelthring*. It is one of the most important and widely used ingredients, and there are around three hundred popular seabuckthorn-based formulations in various pharmacopeias of *Sowa Rigpa*. Seabuckthorn is mixed with up to 50 other herbs or substances to create the formulations (Gurmet, 2009).

sTar-Bu is the classical name for seabuckthorn in *Sowa Rigpa*. The plant is also referred to as *gLawa-tsirma*, *Lai-tsir*, *Tsar-tsir*, *Mai-ldan*, *Stin-ge-sochan*, *Yalga-*

gyur. The berries are the most widely used parts in medicine.

In Tibetan medicine, the three main species of seabuckthorn have been identified as *Sa-sTar* for *H. tibetana*, *Bar-sTar* for *H. rhamnoides*, and *Nam-sTar* for *H. salicifolia*. The species have been classified based on their height. The 'Sa' means 'ground,' 'Bar' means 'middle,' while 'Nam' means 'sky.' There are minor differences in medicinal uses and potencies among the three species. However, *H. rhamnoides* is the most commonly used species in Tibetan medicine. *H. salicifolia* and *H. tibetana* are considered more effective for treating lung and throat disorders, while *H. rhamnoides* is considered better for treating blood disorders (Gurmet, 2009).

Some seabuckthorn formulations and therapeutic uses in the Tibetan system of medicine are as follows (Gurmet, 2009):

Chang khog rTsa shal: Against wounds, tumors, neurological problems

Chong- bZi 21: Peptic ulcer, acidity, bilious vomiting, diarrhea, phlegm disorders

Dai-Nyug: Appetite stimulant, peptic and duodenal ulcer, bleeding peptic ulcer, indigestion, and stomach problems

Dud rTsi chi sos: Chronic cough and cold, bronchitis

Dud rTsi ril kar: Digestion stimulant, acute colicky pain

Gag sail rgyadpa: Chronic constipation, problem of delivery and placenta

Gar-nag 10: Indigestion, gastric neoplasm, gallstone, gastrointestinal polyps, uterine polyps

Gun dum 10: Bronchitis

Gyam-tsa 10: Spleen disease

Kar po 6 sByor: Irritation in rectum

Mo nad chi Joms chen mo: Gynecological problems

Nul-byor: Edema

Ol-Si 25: Menstrual problems

Or-sman dai chaid: Asthama, stomach disorder, liver disorder, anemia, hiccups, edema, piles

Si-dru 8: Antacid, peptic ulcer, and gastritis

Skad sman jamling drug da: Problem in vocal cord

Skuru-25: Hypertension, blood disorders, fever, acidity, liver and stomach problem

sMug-po nYoms ldan jor-wa: Chronic peptic and duodenal ulcer, acidity and phlegm

sMug-poi-ninpo-bDun-sByor: Peptic ulcer and antacid

sNon boo 6 thang: Irritation of skin, eczema and other skin problems

Srolo 11: Asthma, stomach and liver disorder, anemia, hiccups, edema and piles.

sTar-bu 18: Cervicitis, vaginitis, salpingitis, menstrual disorder

sTar-bu 19: Gynecological problems like vaginitis, salpingitis

sTar-bu 3 sByor: Digestive stimulant and against neoplasm

sTar-bu 5: Throat infections

sTar-bu pan tsa: Against vomiting and bronchitis.

sTar-chang: Traumatism and anti-tussive

Tan-joms Zla sil: Gallstone, blood tumor

Tan-nad chi-sman: Tumors, gastric neoplasm, gallstone, gastrointestinal polyps, uterine polyps

Udpal 25: Used against poison, peptic ulcer, liver disorder, gall disorder

Udpal 8: Liver and blood disorder

Usu 6: Gastric, diarrhea, dysentery

Zi-ched 11: Reduce labor pain, dissolve hematoma

Zla shail dud rtsi thigpa: Chronic fever, dysentery, blood vomiting, edema

In the Ladakh region, *Amchi* (traditional doctors) still prescribe preparations made from seabuckthorn to treat common health issues such as indigestion, throat infections, gynaecological problems, ulcers, gastritis, bronchitis, acidity, diarrhea, hypertension, blood disorders, fever, tumors, gallstones, coughs, colds, and food poisoning.

Experimental Studies in Human Subjects

The use of plants for medicinal purposes dates back to ancient times and is considered the foundation of modern medicine. Many of the drugs used today originate from plant extracts. However, to establish the safety and effectiveness of plant-based medicines, they need to undergo clinical trials on human subjects. Researchers have conducted several clinical trials to understand the potential of seabuckthorn extracts in treating various diseases and have gained valuable insights into their safety and efficacy.

Antihypertensive: Clinical observations suggest that the total flavones extracted from seabuckthorn fruits and leaves have antihypertensive effects (Xu et al., 2001). Long-term consumption of seabuckthorn puree (90 ml per day) shows antihypertensive effects in hypercholesterolemic patients. The diastolic blood pressure decreased from 82.3 to 77.8 mm Hg 12 weeks

after puree consumption (Zhou et al., 2022). Supplementation of seabuckthorn seed oil at a daily dose of 0.75 ml for 30 days lowers blood pressure in young male hypertensive human volunteers at high altitudes (Vashishtha et al., 2017).

Cardiovascular disease: Supplemental seabuckthorn improves blood lipid profiles in subjects with higher cardiovascular risks. There is compelling evidence that including seabuckthorn in one's diet can positively impact clinically relevant blood lipid profiles in individuals with hyperlipidemia. Subjects given supplemental seabuckthorn in their diet showed a significant reduction in total cholesterol, triacylglycerol, and LDL-cholesterol concentration in serum/plasma. The cardioprotective effects of seabuckthorn intake may be attributed to its phytochemical constituents, especially flavonoids and beta-sitosterol (Guo et al., 2017).

The effectiveness of seabuckthorn seed oil in reducing dyslipidemia, hypertension, and cardiovascular risk factors in humans has been demonstrated. Dietary supplementation of seabuckthorn seed oil at a daily dose of 0.75 ml for 30 days to hypertensive human subjects reduces serum triglycerides and LDL-cholesterol. It also results in the normalization of blood pressure and reduction in circulatory cholesterol. The effects of the oil on human subjects with normal blood pressure and cholesterol are less pronounced (Vashishtha et al., 2017).

Hepatoprotective activity: Nonalcoholic fatty liver disease (NAFLD) is one of the most common causes of chronic liver injuries in the world. There is no definitive therapeutic strategy for effective treatment of NAFLD. Forty-six NAFLD patients showed promising results when given seabuckthorn capsules at a daily dose of 1.5 g thrice a day for 90 consecutive days. The blood triglycerides decreased from 2.26 to 1.74 mmol/L, and the low-density lipoprotein cholesterol level decreased from 3.2 to 2.6 mmol/L after three months. Similarly, the serum alanine aminotransferase decreased significantly from 69.6 to 50.1 IU/L. The liver stiffness measurement reduced considerably, and the CT liver/spleen ratio increased three months after the treatment (Gao et al., 2014).

Alleviate functional dyspepsia symptoms: Functional dyspepsia represents a highly significant public health issue. Unexplained fullness after eating and an inability to finish a normal-sized meal are common symptoms. Seabuckthorn emulsion (juice and oil), when given at a daily dose of 5-15 g to children with functional dyspepsia for eight weeks, increases the level of appetite factors, leptin, and neuropeptide Y, gastric emptying and gastrointestinal digestive function, and children's growth and development (Xiao et al., 2013).

Burn healing: Burn is one of the most severe forms of trauma that can result in disabilities, fatalities, and mental health issues. Seabuckthorn is effective in

accelerating the burn healing process. A triple-blind clinical trial was conducted on 55 patients with second-degree burns. Half the patients were treated with seabuckthorn ointment and the other half with silver sulfadiazine ointment daily until the burns healed completely. Patients treated with seabuckthorn ointment healed in 6.7 days, whereas those treated with silver sulfadiazine ointment healed in 11.2 days (Shafiee et al., 2021).

Beneficial effects on vaginal atrophy in postmenopausal women: Vaginal atrophy is the thinning and drying of the vaginal mucosa. Typical symptoms include vaginal discomfort, feelings of dryness, burning, and itching. The standard treatment for vaginal atrophy is the use of estrogen hormone, either topically or via a systemic route. Though effective, many women are not willing to use hormone therapy due to its association with a higher risk of breast and endometrial cancer. A dietary supplement of seabuckthorn oil at a daily dose of 3 g for postmenopausal women shows beneficial effects on vaginal health. Therefore, seabuckthorn can be a potential alternative for mucosal integrity for those women not able to use estrogen treatment for vaginal atrophy (Larmo et al., 2014).

Atopic dermatitis: The characteristic signs of atopic dermatitis include dry, scaly, itchy skin with eczematous inflammation and a typical distribution of lesions. Traditionally, therapy for this condition

includes using topical emollients, avoiding skin irritating factors, applying topical glucocorticoids, taking systemically administered antihistamines, and using antimicrobials. Efforts are made to find alternative treatments, including dietary management, due to side effects of steroids and certain antihistamines. During a four-month study, patients with atopic dermatitis were investigated for the effects of dietary supplements containing seabuckthorn seed and pulp oil. Supplementation with seed oil increased linoleic and alpha-linolenic fatty acid in patient plasma lipids, resulting in a beneficial effect on atopic dermatitis (Yang et al., 1999).

Reduce hyperpigmentation: Hyperpigmentation is one of the most common skin disorders. An increase in melanin synthesis results in hyperpigmentation or spots, especially on the face and neck. A study on 25 melasma patients found that applying seabuckthorn-based emulsion formulation reduces skin melanin. A 3.5% reduction in skin melanin was observed after one week of treatment, and a 16.35% reduction after 12 weeks (Khan et al., 2013).

Anti-obesity: Obesity is a serious health issue. It is the root cause of many disorders, including diabetes, cardiovascular diseases, cancer, fatty liver, hypertension, cardiac hypertrophy, and decreased life span. Seabuckthorn acts as an anti-obesity agent. A capsule containing hydroalcoholic extracts of seabuckthorn, *Dioscorea bulbifera*, and *Terminalia*

chebula was given to 65 obese adolescents daily for three months. A considerable improvement in lipid profile levels, inflammatory cytokines, adipokines, and blood pressure was observed after the treatment. A decrease in BMI and hip-to-waist ratio was recorded compared to an equal number of patients who received placebo capsules (Shivakumar et al., 2015).

In another study with 41 obese children, intake of seabuckthorn pulp oil (800 mg per day) for 60 days prevented metabolic syndrome in hypertriglyceridemic waist phenotype obese children by reducing triglycerides and blood pressure. The treatment resulted in a 37.5% reduction in triglycerides (Casariu et al., 2013).

In a sugar challenge test on 18 obese male subjects, seabuckthorn fruit was found to decrease and delay insulin response. The study suggests using seabuckthorn as a culinary tool for lowering meal insulin response (Mortensen et al., 2018).

Dry eye: Dry eye is a common condition that causes severe discomfort and dramatically compromises the quality of life. Artificial tears are a standard treatment for dry eyes, but they do not solve the condition's root cause and the inflammation accompanying dry eyes. Anti-inflammatory drugs relieve the inflammation but are unsuitable for long-term use and may cause side effects. A three-month dietary supplement with seabuckthorn seed and pulp oil attenuates the

increase in tear film osmolarity during the cold season. It also positively influences the maximum intensity of redness and burning symptoms in patients with dry eyes (Larmo et al., 2010).

Blood clotting: A study conducted on 11 healthy normolipidemic men showed that consuming seabuckthorn seed and pulp oil (5 g per day) for four weeks is beneficial in treating persons with an increased tendency to blood clotting. Seabuckthorn oil supplementation reduces both the velocity and total platelet aggregation (Johansson et al., 2000).

Experimental Studies in Animal Model

The discovery and development of any given drug are invariably lengthy and expensive processes in the pharmaceutical industry. Animal models are used to establish and provide evidence for non-clinical proof-of-concept for the safety, efficacy, and target of interest of specific drugs for subsequent human use. Seabuckthorn-based therapeutic agents have been studied on animal models for safety and effectiveness in treating certain diseases.

Protection against arsenic toxicity: Inorganic arsenic is found in soil and water in various locations. An estimated 200 million people are chronically exposed to inorganic arsenic through contaminated drinking water. Arsenic is also utilized in multiple industries, such as pesticides, wood preservatives, glass

manufacturing, and semiconductor materials. Exposure to arsenic over a long period can cause several human diseases. Researchers have discovered that a seabuckthorn fruit extract significantly protects against arsenic-induced oxidative injury in mice (Gupta & Flora, 2006).

Neuroprotection against iron-induced epilepsy:

Epilepsy is a neurological disorder associated with many forms of congenital and acquired brain pathology, such as tumors, stroke, and traumatic brain injury. Administration of seabuckthorn berry extract reduces the epileptiform activities in the cortex and hippocampus of epileptic rats. It also improves memory impairment, reduces anxiety-like behaviors, and improves histological impairments in iron-induced epilepsy (Ladol & Sharma, 2021).

Anti-stress and adaptogenic activity: Herbal formulations that can enhance physical endurance, mental function, and non-specific resistance to stress have been termed adaptogens. Seabuckthorn dry leaf extract administered orally to rats 30 minutes before cold, hypoxia, and restraint exposure shows significant anti-stress and adaptogenic activity with no toxicity. The extract possesses physical performance-enhancing activity when the rats are exposed to stressful environments (Saggu et al., 2007; Saggu & Kumar, 2008).

Cardioprotective effect: The curative effect of seabuckthorn preparation on cardiovascular diseases is well documented. Flavonoids in berries and leaves, as well as oil are well-known to improve the functioning of the cardiovascular system. The total flavonoids from leaves and fruit exhibit protective effects on myocardial ischemia (Eccleston et al., 2002).

Seabuckthorn oil has been demonstrated to mitigate myocardial damage in isoproterenol-induced cardiac injury in rats by maintaining hemodynamic, biochemical, histopathology, and ultrastructural perturbation owing to its free radical scavenging and antioxidant activities. Therefore, a diet containing seabuckthorn oil may benefit those at risk of myocardial injury (Malik et al., 2011).

Hypertension provoked by a high-sucrose diet has been found to be alleviated or modulated by total flavones purified from the seabuckthorn seed residue. Feeding rats a high-sucrose diet for six weeks increases systolic blood pressure by 25.6%. However, treatment of the rats with total flavones significantly suppressed the elevated hypertension two weeks after the treatment. The antihypertensive effect is partly attributed to improvement in insulin sensitivity and blocking the angiotensin signal pathway (Pang et al., 2008).

Anti-psoriatic activity: Psoriasis is a chronic inflammatory skin disease characterized by red

plaques with silvery-white scales associated with itching, burning, and irritation. Currently, there are no permanent treatments for psoriasis, and anti-inflammatory drugs are topically applied during the flaring of the disease. In vivo oral and topical treatment with seabuckthorn pulp oil shows anti-psoriasis-like efficacies in healing chemical-induced psoriasis in mice. The oil can be used as a nutraceutical with anti-psoriatic properties (Balkrishna et al., 2019).

Antiulcer activity: Seabuckthorn seed and pulp oil have protective and curative effects on gastric ulcers. Oral administration of the oil significantly reduced the ulcer formation in rats. The antiulcer activity is attributed to high levels of β -carotene, α -tocopherols, and β -sitosterol in the oil. The oil also inhibits lipid peroxidation in gastric mucosa (Xing et al., 2002; Krichkovskaya et al., 1992).

Protection against chemical warfare agent: Sulphur mustard, commonly known as mustard gas, is an alkylating agent that causes serious blisters upon contact with human skin. It was first used in 1917 and has been used as a warfare agent in several instances since then. Seabuckthorn leaf extracts and flavones from the fruit have shown a protective effect against the warfare agent in mice (Vijayaraghvan et al., 2006).

Hepatoprotective activity: The liver can be damaged by environmental pollutants and drugs, leading to hepatitis or cirrhosis. Seabuckthorn leaf extract and

seed oil ameliorate liver injury in animal models. Oil administered orally is comparable with silymarin in protecting mice against CCl₄-induced hepatotoxicity. The hepatoprotective effect is attributed to several constituents with potentially healthy biological properties, such as α -tocopherol and β -carotene (Geeta et al., 2009; Hsu et al., 2009).

Antidepressant properties: Depression is a chronic illness that affects an individual's mood, thoughts, physical health, and behavior. However, synthetic antidepressants often cause side effects like dry mouth, constipation, sexual dysfunction, and an inability to drive skillfully. To manage stress and depression, people are searching for plant-based formulations. Oral administration of seabuckthorn aqueous fruit extract to experimental rats for two weeks exhibits antidepressant-like properties. The results suggest that seabuckthorn fruit extract is effective as an antidepressant plant material (Batool et al., 2011).

Anti-obesity: Seabuckthorn flavone-enriched seed extract exhibits significant anti-obesity, triglyceride-lowering, and hypoglycemic effects in high-fat diet-induced obese mice. Administration of the extract for nine weeks at 100 and 300 mg per kg body weight significantly reduces the body weight gain by 33.1 and 43.5%, respectively. The triglyceride concentration in serum and liver decreases by 15.7 and 49.6% (Yang et al., 2017). The leaf extract also has a beneficial effect in

preventing weight gain in high-fat diet-induced obese mice. The extract efficiently prevents an increase in hepatic triglycerides and cholesterol (Pichiah et al., 2012).

Anticancer activity: Current therapies against cancer are still unsatisfactory, mainly because of their side effects. In recent years, natural polysaccharides that enhance host defense have been recognized as a possible means of inhibiting tumor growth with low toxicity and few side effects. A polysaccharide (HRWP-A) isolated from seabuckthorn berries shows antitumor activity in Lewis lung carcinoma tumor growth in mice. At day 16, the tumor volume in mice treated at 200 mg per kg body weight of HRWP-A was comparable to that in the positive control treated with cyclophosphamide (CTX). The findings suggested that HRWP-A inhibited tumor growth by enhancing innate immune response (Wang et al., 2015).

Methanol extracts of seabuckthorn bark also show anticancer activity in mice transplanted with tumor cells. A significant reduction in tumor volume, weight, and viable cells were observed in mice treated with the extract. The mice treated with the extract survived for 43 days, compared to only 21 days for the mice with tumor cells (Chakraborty et al., 2015).

Experimental Studies in Cell Lines

The discovery and development of effective medicines have historically been hampered by the need for more

reliably predictive preclinical models to assess the therapeutic efficacy of candidate agents. The cell line-based platform remains the best and possibly the only tool for the identification and characterization of medicinal agents that can potentially produce clinical benefits, especially in cancer patients. Seabuckthorn extracts have been used to study their therapeutic potential in cell lines.

Anticancer: Breast cancer is a heterogeneous disease caused by genetic and environmental factors and is the leading cause of cancer death in women worldwide. Seabuckthorn fruit extract is known to inhibit the proliferation of colon and breast cancer cells. Antioxidants and anthocyanins are attributed to the inhibition of cancer cell proliferation (Olsson et al., 2004). The berry and seed extracts show antitumor activity in breast cancer cell lines. The inhibition of cancer cell proliferation and apoptosis induction by the berry extract is attributed primarily to high carotenoid contents (Visan et al., 2023).

Isorhamnetin, a flavonol aglycone isolated from seabuckthorn, shows antitumor activities in human carcinoma cells. Isorhamnetin exerted cytotoxicity against human hepatocellular carcinoma cells and induced cell cycle arrest at the G₁ phase (Teng et al., 2006). Fatty acid synthase is overexpressed in many human cancer cells and is considered a promising therapy target. Procyanidins isolated from seabuckthorn seed have an inhibitory effect on fatty

acid synthase activities in human breast cancer cells (Wang et al., 2014).

Protection against ultraviolet radiation: Exposure to solar ultraviolet (UV) radiation, particularly UVB, has various effects on the skin, including sunburn, erythema, hyperplasia, photoaging, hyperpigmentation, and immunosuppression. A low dose of UVB can cause DNA mutation, while high doses can result in irreparable DNA damage, leading to apoptosis. Seabuckthorn seed extract exhibits a significant photo-protective effect against UVB-induced cytotoxicity in cultured human skin dermal fibroblast. Therefore, the seed extract has potential applications in preventing and treating skin photoaging (Kim et al., 2012).

Therapeutic Applications in Veterinary

Seabuckthorn has been used for centuries to treat various animal diseases. Recent clinical trials have confirmed the plant's effectiveness in treating multiple veterinary ailments.

Wound healing: Seabuckthorn has therapeutic properties in healing infected cutaneous wounds in bovines. An ointment containing seabuckthorn fruit powder exhibits better wound contraction than treatment with povidone-iodine ointment (Mahajan et al., 2006).

Gastric ulcerations and erosions: Gastric ulcerations and erosions (GUEs) in small animals develop mainly due to long-term administration of steroidal and non-steroidal anti-inflammatory drugs. The drugs used for the management of GUE have many side effects, and thus, people are looking for alternate medicines. Oral administration of seabuckthorn seed oil (5 ml per animal) twice a day to dexamethasone-induced GUE in dogs shows better therapeutic efficacy than standard drugs such as famotidine, lansoprazole, misoprostol, and sucralfate (Dogra et al., 2013).

Seabuckthorn has a fascinating medicinal history that goes back to the Greeks and Romans. The medicinal benefits of the shrub are well documented in the 8th-century Tibetan medicinal classic rGyud Bzi. Current research is now beginning to understand and support the traditional uses. Clinical studies have shown the effectiveness of seabuckthorn-based formulations in reducing dyslipidemia, hypertension, and cardiovascular risk factors. Cell line studies have shown the potential of seabuckthorn extracts in treating cancer and in preventing and treating skin photoaging.

Chapter-5

Seabuckthorn in the Cosmetic Industry



Seabuckthorn seed and berry oil from Ladakh

A cosmetic product refers to any substance intended to come into contact with the external parts of the human body to clean, protect, perfume, or enhance their appeal. In recent years, the term 'cosmeceuticals' has been widely used to describe cosmetic products that demonstrate some pharmaceutical activity, usually by containing natural ingredients with

beneficial results. North America is a major natural and organic personal care product market, followed by Europe and Asia-Pacific, where China and India play a significant role in the global herbal cosmetics market. The global consumption of plant-based cosmetics has shown spectacular growth in recent years due to increasing consumer awareness of the benefits of naturally derived ingredients that have traditionally been used.

Seabuckthorn, particularly its oil, is widely used in the cosmetic industry to produce various products, including skin care creams, hair shampoos, body lotions, and beauty creams. It is primarily found in anti-aging and anti-wrinkle products. Applying the oil directly to the skin can improve the skin tone, giving it a fresh and healthy appearance (Zielińska & Nowak, 2017). The oil has been verified to have well-balanced content of fatty acids, carotenoids, and vitamins, which helps in retarding skin aging, absorbing harmful irradiation, and protecting skin against inflammations and bacteria growth (Zosimidou et al., 2023).

Studies have shown that seabuckthorn leaf and fruit extracts have high antioxidant activity, which effectively eliminates free radicals, enhances cellular antioxidant capacity, and reduces environmental damage to the skin. Adding seabuckthorn extract to cosmetics significantly enhances the antioxidant and anti-aging functions of the product. The efficacy and safety of seabuckthorn extracts in cosmetics have

been confirmed through human skin application testing (Wan et al., 2024).

Seed oil for healthy skin

Linoleic acid (18:2) is the primary fatty acid in seabuckthorn seed oil (up to 42%) and is also present in the berry and pulp oil in lower concentration levels (6-33%). It is also known as omega-6 fatty acid. Together with α -linolenic acid (ALA), linoleic acid constitutes the essential fatty acids that human body cannot synthesize. It is the most abundant polyunsaturated fatty acid in human skin, and dietary deficiency in linoleic acid results in a characteristic scaly skin disorder and excessive epidermal water loss (Solà Marsiñach & Cuenca, 2019).

The lamellar granules of human skin produce lipids that maintain the skin's protective barrier to avoid epidermal water loss. The reproduction of lamellar granules slows down with aging, and the skin becomes drier and weaker (Zielińska & Nowak, 2014). Omega-6 might reverse this process by accelerating the production of lipids in lamellar granules, strengthening the lipid barrier of the epidermis, protecting the skin against epidermal water loss, and normalizing the skin's metabolism (Catalán et al., 2016). In addition, people with acne skin present a decrease of the fatty acid in sebum, which leads to blocked pores and the formation of comedos and eczema. In addition, it is thought that omega-6 might improve

sebaceous gland activity, unblocking pores and decreasing the number of comedos (Zielińska & Nowak, 2014). Therefore, topical application of facial cream containing seabuckthorn seed oil results in healthy skin.

Anti-aging and anti-wrinkle

As a powerful antioxidant, seabuckthorn oil may delay aging by removing free radicals. It is added to anti-aging and anti-wrinkle cosmetics (Zielińska & Nowak, 2014). The properties of the berry oil in reducing skin wrinkles, blemishes, and pigmentation and improving facial cheek lifting, skin elasticity, density, and skin tone have been verified in women aged 30-60 years old. Applying the oil and hot-water extracts for six weeks significantly reduces wrinkles around the eyes (12.9%) and spots and pigmentation (2.5% improvement). Its application improves facial cheek lifting (44.6%), skin elasticity (4.3%), skin density (9.6%), and skin tone (1.6% improvement) (Lee et al., 2021). Blending seabuckthorn fruit extract with *Rubus fruticosus* and *Perillae folium* leaf extracts in a 5:14:1 ratio shows a 28% increase in elastase inhibitory effect, indicating wrinkle improvement (Kim et al., 2019).

Reduce hyperpigmentation

Hyperpigmentation is one of the most common skin disorders. An increase in melanin synthesis results in

hyperpigmentation or spots, especially on the face and neck. Applying seabuckthorn-based emulsion formulation reduces skin melanin. A 3.5% reduction in skin melanin is reported after one week of treatment, and a 16.35% reduction after 12 weeks (Khan et al., 2013). Therefore, topical application of seabuckthorn oil reduces hyperpigmentation. The blending of seabuckthorn fruit extract with leaf extracts of *Rubus fruticosus* and *Perillae folium* in a 5:14:1 ratio shows a 24.4% improvement in tyrosinase inhibitory activity, which is a whitening index (Kim et al., 2019).

UV-protection

The skin is constantly exposed to numerous external factors that can impair skin functioning as a natural biological barrier, including ultraviolet (UV) radiation in sunlight. High-energy UV radiation absorption by the skin may trigger mechanisms that defend skin integrity and induce skin pathology, such as cancer. Seabuckthorn seed oil prevents UV-induced disturbances in redox balance and lipid metabolism in skin fibroblasts and keratinocytes, indicating it is a promising natural compound in skin photoprotection (Gęgotek et al., 2018). Russian cosmonauts used seabuckthorn oil to protect against harmful radiation.

Atopic dermatitis

The characteristic signs of atopic dermatitis include dry, scaly, itchy skin with eczematous inflammation

and a typical distribution of lesions. Dietary supplements containing seabuckthorn seed and pulp oil given to patients with atopic dermatitis increase linoleic and alpha-linolenic fatty acids, which has a beneficial effect on atopic dermatitis (Yang et al., 1999).

Hair strengthening

Seabuckthorn strengthens hair, so it has been used in shampoos, conditioners, and other hair products. These products target the recovery of damaged hair, elasticity restoration, and hair loss prevention (Koskovac et al., 2017).

The increasing demand for plant-based cosmetics highlights a growing consumer awareness of the benefits of naturally derived ingredients. The use of seabuckthorn oil by the Russian cosmonauts during the space race of the 1960s and 1970s is the first sign of research-based findings on using seabuckthorn in the cosmetic industry. Scientific research has shown that the oil contains a well-balanced combination of fatty acids, carotenoids, and vitamins, which can help slow skin aging, protect skin from harmful radiation, and prevent inflammation. Furthermore, studies have confirmed that the oil can effectively reduce skin wrinkles, blemishes, and pigmentation. Human skin application testing has also confirmed the safety and effectiveness of this berry oil in cosmetic applications.

Chapter-6

Seabuckthorn in the Food Industry



Seabuckthorn-based products from Ladakh

Seabuckthorn is a highly valued raw material in the food industry, especially for functional foods and dietary supplements. While all parts have diverse applications, the fruit pulp and seed are the most valuable components. Seabuckthorn has been used to create more than 200 products in different parts of the world. As the number of research and development centers and seabuckthorn-based industries continue to grow, the list of products is expected to expand even further. China alone has more than 1,500 registered companies producing food products from

seabuckthorn (Chen et al., 2023). The products are available in many countries, including Canada, China, Finland, Germany, India, and the United States.

Seabuckthorn has several advantages over other fruit crops in the food industry. The plant has an attractive nutritional mystique relating to its thousand-year history, which dates back to the Greeks and Romans. Seabuckthorn berries are among the most nutritious and vitamin-rich of all fruits. It is also rich in various antioxidant compounds. Consumers can be offered healthy products without any added supplements. Scientific research has shown that the plant's claimed nutritional value has increasingly solid support from the scientific community. Seabuckthorn's market value is generated primarily by value-added processed products. Therefore, these products have long shelf lives and are not limited to a particular season or region (Li & Beveridge, 2007).

Seabuckthorn is a fruit that is not affected by overproduction. Because of this, it is considered a premium product and is sold at high prices when fresh or processed. The shrub grows in land unsuitable for other crops and is typically limited to certain regions with specific ecological conditions. It adapts well to relatively dry, sterile soil, with excellent winter hardiness and an inherent toughness, where most other plant species cannot thrive (Li & Beveridge, 2007). The principal source of the berries has been areas planted with seabuckthorn for land reclamation

and erosion control. Such harvests provide a unique advantage, particularly in terms of cost of production and wild harvest. Seabuckthorn is already a well-established crop in Asia; millions of hectares are under cultivation.

Juice Products

Seabuckthorn berry is highly acidic, with a pH of 2.7-3.1 (Seglina et al., 2006). Therefore, the juice extracted from the berries cannot be consumed directly. However, the pulp is nutrient-rich and is processed into various products. Freshly pressed juice is a complex mixture of fluid, suspended solids (cloud), and oil, and is, therefore, a three-phase system that requires stabilization for most acceptable products. A combination of emulsifiers, natural hydrocolloid stabilizers, and homogenization retard phase separation and gives a product with good stability (Beveridge et al., 1999). However, consumers who are well aware of the natural characteristics of the juice generally prefer juice products without any added emulsifiers and hydrocolloids. Coloring and flavoring agents are generally not used. The juice is highly nutritious. Russian cosmonauts were supplied with seabuckthorn beverages to enhance their health and resistance to stress. It was the first fruit juice in space. The seabuckthorn drink was the official beverage for Chinese athletes at the Seoul Olympic Games in 1988 (Small et al., 2002).

Ready-to-serve: Ready-to-serve beverages are non-fermented drinks that can be consumed directly after mixing pulp, sugar, water, and additives. These drinks contain up to 10% fruit juice. Sometimes, seabuckthorn pulp is blended with other fruit juice in varying proportions to balance the sensory characteristics and reduce the cost.

Squash: Squashes are concentrated fruit juices containing 25% fruit pulp. They are consumed by diluting them with water, as they are too strong. The beverage is quite popular among consumers. Seabuckthorn pulp is often mixed with other fruit juices to enhance the sensory characteristics of the drink. One example is Seapricot, made by blending seabuckthorn and apricot juice.

Pulp: Pulp, with or without sugar, is popular among health-conscious consumers. The product label generally declares that pulp separation is a natural phenomenon, and consumers are advised to shake the bottle well before use. Pulp is consumed by mixing 2-3 tablespoons of the pulp in a glass of water. Such products have up to an 18-month shelf life.

Fermented Juice: Seabuckthorn juice is known for its sourness caused by organic acids, which can be a disadvantage. However, fermentation of the juice for 18 hours can reduce the sourness and astringency while increasing the sweetness. This process also enhances the fruity flavor of the juice by producing

esters during fermentation. As a result, fermented seabuckthorn juice is more flavorful and appealing (Tiitinen et al., 2007).

Jam and Jellies

Jam and jellies contain 45% fruit pulp. Seabuckthorn juice is generally blended with other fruit juices, such as apricots, grapes, apples, strawberries, raspberries, and currants, to improve its sensory characteristics. These products are highly nutritious and can be stored for up to 6 months at room temperature.

Wine

During Genghis Khan's rule, Mongolians were the first to prepare seabuckthorn wine from highly acidic berries. This wine was exclusively served to the royal families and nobles (Chen et al., 2023). Seabuckthorn wine has a high phenolic content, which makes it potentially beneficial for health, more so than fruit wine made from grapes, black currant, and apples (Negi & Dey, 2009). The wine has a golden color and a pleasant aroma. It is a popular product in the Czech Republic. Seabuckthorn is also used as a raw material in beer production (Lee et al., 2012).

Herbal Tea

Seabuckthorn leaf is used as a major constituent in formulating herbal tea. Shade-dried leaves are mixed with other high-altitude herbs like salam panja

(*Dactylorhiza hatigeria*), tea (*Bidens pilosa*), caraway (*Carum carvi*), black caraway (*Bunium persicum*), oregano (*Origanum vulgare*), mint (*Mentha longifolia*), yarrow (*Achillea* sp.) and rose root (*Rhodiola* sp) to improve the sensory characteristics. The tea is rich in antioxidants, and the product is commercially available.

Pigment

The yellow-orange pigment is extracted as a byproduct from the residue after juice extraction. It contains flavones, carotene, and vitamin E. The pigment has potential applications as a coloring additive in pharmaceutical, cosmetic, and food industries where yellow-orange colors are desirable.

Dairy Products

Seabuckthorn promotes the growth and reproduction of probiotics. Therefore, seabuckthorn is combined with probiotics to provide functional characteristics to dairy products. Using dry seabuckthorn berries as an immobilization carrier for the probiotic strain in feta cheese results in a soft and smooth taste and positively contributes to the aroma profile. The sour taste of the berries is masked by the cheese's strong flavor (Terpou et al., 2017). The berry is also used as a probiotic cell immobilization carrier to develop functional frozen yogurt. Immobilization of the strains improved their survival rate during storage and protected the probiotic against gastric conditions. It

also enhances the overall acceptability of the yogurt (Terpou et al., 2019). Therefore, adding seabuckthorn improves dairy products' sensory, physical, and functional characteristics.

Bakery Products

Most bakery products are made using refined flour. This type of flour has good baking properties but lacks essential vitamins, minerals, and dietary fibres. However, using seabuckthorn powder in bakery products can significantly increase their antioxidant activity, resulting in healthier products. Research has shown that adding 3% peel-based or 5% seed-based powder to bakery products yields the best results in terms of product quality. Products baked at 200°C have more antioxidants than those baked at 220°C (Nilova & Malyutenkova, 2018). Biscuits containing a specific amount of seabuckthorn pulp and leaves have been developed. These biscuits are a healthy snack option due to their high dietary fibre, flavonoids, and other phytochemicals. In addition, other bakery products, including buns, bread, rusks, and cakes, have also been created using antioxidant-rich seabuckthorn leaves.

Juice Powder

The liquid form of seabuckthorn juice often makes it challenging to incorporate into dry food products. One way to overcome this difficulty is spray drying, which converts the juice into dry powder while

preserving its natural color, nutrients, and flavor. Another method is vacuum freeze-drying, which retains even more nutrients. The resulting powder can be used to create other products like pulp, wine, and more.

Products developed from seabuckthorn are considered premium and sold at high prices. The plant's claimed nutritional, medicinal, and cosmetic values have increasingly solid support from the scientific community. Over 200 products have been developed, and the list will expand further. Several products marketed as seabuckthorn-based contain very little seabuckthorn. Consumers must carefully select the right products to benefit from the wonder plant.

Chapter-7

Environmental Benefits of Growing Seabuckthorn



Thorny and bushy growth provides a protective shelter for flora and fauna

Soil erosion is a significant environmental issue in mountainous areas. Communities that suffer from severe soil and water loss face devastating consequences for agriculture, animal husbandry, forestry, and ecology. Farmers face a shortage of firewood for 3-5 months annually. In northern and northwestern China, extensive tree and shrub planting

has been carried out since the 1950s to control soil and water loss caused by environmental degradation. Planting certain species, including Chinese pine, poplar, and elm, has effectively controlled environmental degradation in some areas. In the 1970s, some species were planted on the Loess Plateau using air-seeding techniques. However, many parts of the country could not replicate the success stories despite government persuasion and subsidies. Farmers often complain that they do not get direct economic benefits from planting such trees. When the knowledge about seabuckthorn's rich nutrient contents and medicinal values reached China from the Soviet Union in the 1980s, they became aware of the shrub's tremendous economic and ecological benefits. Seabuckthorn was then extensively cultivated throughout China's northern, northwestern, and southwestern parts. A natural seabuckthorn forest can yield 0.75-1.5 tons of berries per hectare, which has huge economic value. One hectare of seabuckthorn plantation yields 4.5 tons of dry firewood and 1.5 tons of leaves (Rongsen, 1992). The seabuckthorn plantation has become a massive success due to its economic benefit and multipurpose usage in controlling environmental degradation.

Growing seabuckthorn in the marginal soil of the fragile ecosystem has potential applications in soil and water conservation, land reclamation, desertification control, and establishment of wildlife habitats, besides social and economic benefits. Due to its drought

resistance, soil salinity tolerance, and cold weather tolerance, the shrub is suitable for many demanding environments where most plants wouldn't survive. Several parts of the world have witnessed successful cases of environmental restoration.

The shrub develops an extensive root system, making it ideal for soil binding. The shrub spreads rapidly through the suckers, which develops a few years after planting. Root nodules of seabuckthorn can fix atmospheric nitrogen. Additionally, the roots can transform insoluble organic matter in the soil into more soluble forms, which are essential for plant growth and development.

Seabuckthorn windbreaks prevent wind erosion in open areas. Plants that serve as windbreaks must resist the drying effects and physical injuries caused by wind, and seabuckthorn is well suited to this task. The shrub's survival rate in forest areas is over 80%, and thus, it survives more easily than other trees and bushes (Rongsen, 1992).

The seed germination rate is high, and the seedlings grow fast. The young plants develop strong root systems. These features make seabuckthorn suitable for air-seeding. Since the late 1950s, seabuckthorn afforestation through air-seeding has been carried out in the mountain areas of northwestern China. It's a promising method where only 0.4 to 0.5 kg seeds are required per hectare. The cost of air seeding is one-

third that of conventional afforestation. A small airplane can do air-seeding on over 1,300 hectares in a day. The seedling survival rate ranges from 1-25%, and 60% die during the following years. An average of one plant per 10 m² survive after air-seeding. However, after the third year, an increase in the number of seabuckthorn plants is observed since new plants begin to emerge from the roots. Air-seeding, therefore, is an effective method to accelerate the control over soil and water loss. The major constraint in air-seeding is the rainfall after seeding. The seed germinates well when there are 6-8 continuous cloudy days and more than 50-60 mm of precipitation (Rongsen, 1992).

Soil Fertility Enhancement

Root nodules of seabuckthorn have a symbiotic association with a bacterium belonging to the genus *Frankia*, which can fix nitrogen in non-leguminous woody plant species, including seabuckthorn. Seabuckthorn fixes 180 kg of nitrogen per hectare per year, improving soil fertility (Jike and Xiaoming, 1992). In the Loess Plateau in China, a 6-year-old seabuckthorn plantation contains about 115% more soil organic matter and 90% more nitrogen than the soil without seabuckthorn (Qinxiao & Hingyan, 2003). Similarly, the available phosphorus and potassium contents in seabuckthorn sites are 30.3 and 90.3% higher than in sandy soil (Yi, 1993).

Desertification Control

Seabuckthorn is an ideal plant for desertification control. The plant has been used successfully in Jianping Country, Loess Plateau, and Inner Mongolia (www.icrts.org). Jianping country has witnessed frequent floods, droughts, and storms due to indiscriminate logging, leading to only 1.9% forest coverage. However, the plantation of seabuckthorn on more than 67,000 hectares of land has created the largest seabuckthorn forest in the world. Similarly, in the Chinese Loess Plateau, environmental degradation covering an area of 430,000 km² has resulted in annual topsoil losses of 1,600 million tons. Search for a suitable plant species in terms of survival rate, multiplication, and soil improvement coupled with economic benefit has resulted in identifying seabuckthorn as the most appropriate species. Seabuckthorn now covers more than 200,000 hectares in the Loess Plateau.

Yike Zhao witnessed large-scale degradation in Inner Mongolia, referring to it as the 'valley of death', 'moon on earth', and 'environmental cancer'. Government initiatives to check environmental degradation have resulted in over 50,000 hectares of land under seabuckthorn. Planting on challenging terrain such as gully slopes, bottoms, elongated mounds, riversides, and rocky areas is successful (www.icrts.org).

Debris Slide Erosion

The debris slide erosion happens due to gravitational force in the gully region. The debris rolls down a steep gradient when the rock is exposed to various weather and developmental activities. In the 1980s, seabuckthorn plantation was used to control such erosion in the Loess Plateau in China (Sheng, 2003).

Soil and Moisture Conservation

Seabuckthorn develops an extensive root system. In a 10-12-year-old shrub, the horizontal root extends up to 5.4 m and vertically up to 1.3 m deep, thus making it an ideal plant for soil binding (Dwivedi et al., 2006). Seabuckthorn plantation intercepts precipitation by the lush crown layer, redistributing precipitation and controlling surface runoff. The adequately developed canopy of 5-7-year-old seabuckthorn stands intercepts about 1-2 kg rainwater per square meter. The canopy also reduced the kinetic energy of the raindrops, thus decreasing soil erosion. Litter production in the 5-7-year-old seabuckthorn stand is around 546 g per square meter, constituting 75.6% of the total water-holding capacity. Litter production also protects the soil surface from running water. Compared to farmland, seabuckthorn stands decreased runoff by 87% and soil loss by 99% (Qinxiao & Hingyan, 2003).



Growing seabuckthorn along water channel to prevent soil erosion

Seabuckthorn plantation improves soil physical characteristics and fertility, thereby improving soil water holding capacity. In the Loess Plateau in China, a 6-year-old seabuckthorn plantation contains about 115% more soil organic matter and 90% more nitrogen than the soil without the seabuckthorn (Qinxiao & Hingyan, 2003).

The soil temperature of seabuckthorn forest remains 3.2°C lower than wasteland (Sheng, 2003). Therefore, it helps in soil moisture conservation. Soil moisture in seabuckthorn plantation areas remains 3-4% higher than outside the forest (Chengjiang & Daiqiong, 2002). Similarly, air humidity is 10-20% higher in seabuckthorn forests. The water surface evaporation

within seabuckthorn forests is reduced by 68.9% compared to wasteland (Sheng, 2003).

Windbreaks and Sand-fixing

Windbreaks made up of seabuckthorn prevent wind erosion in open areas. Plants that serve as windbreaks must resist the drying effects and physical injuries caused by wind, and seabuckthorn is well suited to this task. The spiny shrub has proven beneficial as a barrier to pedestrian traffic, preventing sensitive vegetation from being trampled. The wind velocity at 1 m above the ground surface is reduced by 84.6% due to the seabuckthorn plantation compared to an open area (Sheng, 2003). Planting seabuckthorn is effective in fixing the movement of dunes. It is used extensively in China (Cifen & Guangyao, 2003). The shrub protects the railway lines from the accumulation of sand in the Shandake Desert in China. It also prevents dune movement along the railway lines (Sheng, 2003).

Since the 1960s, the State Forest Department in Ladakh has been expanding the area under seabuckthorn through root suckers, primarily for soil conservation purposes. An area covering more than 70 hectares was planted with seabuckthorn in Hunder village of the Nubra Valley to fix dunes. Recently, plantations have been carried out in Durbuk and Nyoma villages in the Changthang Valley.

Reduces the Harvesting Pressure

Certain regions in the world, such as the cold desert of Ladakh, have limited vegetation and very low temperatures during winter. Therefore, it is necessary to cultivate fast-growing plant species that can be used as firewood for heating homes and cooking food. A single hectare of seabuckthorn forest can produce 10–30 tons of firewood, and the wood can be collected every 4–5 years (Qinxiao & Hongyan, 2003).



A pile of seabuckthorn wood for firewood purpose

Besides, seabuckthorn has proved to be a popular green energy plant because of its quality biomass. The calorific value of dry seabuckthorn is 4785.5 cal per kg. Due to these properties, it has been traditionally used as firewood in Ladakh, India. In monasteries where hundreds of monks reside, the plant species are used

as firewood to cook food. Most monasteries maintain their seabuckthorn growing areas, popularly called *Tsokshing*, to ensure a regular supply of firewood. In the local dialect, *Tsok* means 'forest', and *Shing* stands for 'wood', thus reinforcing the widespread use of seabuckthorn as a firewood plant in the region. The shrub is used as firewood to roast barley grains and prepare the staple food *Tsampa* (Stobdan et al., 2013). Since the shrub grows fast, it reduces the harvesting pressure on native woody plants, thereby maintaining the fragile ecosystem.

Reclamation of Mined Land

Seabuckthorn is extensively used in Russia, Poland, Hungary, and China to return mined land to beneficial end use. The shrub fixes atmospheric nitrogen and accumulates humus in the mined land, thereby increasing soil fertility and the surrounding environment (Sheng, 2003).

Improvement of Microclimate

Seabuckthorn improves the microclimate of the area where it grows. The plant decreases the solar radiation that falls on the ground surface. The canopy reduces solar energy by 41% in summer and 29.6% in winter. It also influences the air temperature. The seabuckthorn forest remains 0.3°C cooler than the surroundings. The shrub also increases the air humidity by 18.9% at the height above the canopy (Sheng, 2003).

Wildlife Habitats

Several wildlife species depend on seabuckthorn stems, leaves, flowers, roots, fruit, and seeds. Thorny and bushy growth provides a protective shelter for flora and fauna, thereby maintaining the fragile ecosystem. In the Loess Plateau of China, 51 bird species are entirely dependent, and 80 are relatively reliant on seabuckthorn for their food (www.icrts.org). In snow-covered winter, the importance of seabuckthorn increases as it is almost the only food available for birds. In the cold arid region of Ladakh, India, the endangered double-humped camel, yak, sheep, goat, etc., survive on the plant mainly during winter when no other fodder is available (Dwivedi et al., 2006).



Goats eating seabuckthorn leaves

One hectare of seabuckthorn plantation can support 10-15 sheep for grazing (Qinxiao & Hongyan, 2003). Seabuckthorn thus provides long-term benefits in maintaining the ecological equilibrium and improving the environment.

Seabuckthorn has the required characteristics to grow on marginal soil. Drought resistance and tolerance to soil salinity and cold weather make the shrub suitable for many demanding environments where most plants fail to thrive. The shrub spreads rapidly through the suckers. Several parts of the world have witnessed successful cases of environmental restoration by planting the wonder plant. Its extensive root system makes it ideal for soil binding and nitrogen fixation. Its thorny and bushy growth provides a protective shelter for flora and fauna.

Chapter-8

Cultivation Practices for Growing Seabuckthorn



Seabuckthorn can withstand -43°C

There is a growing interest in cultivating seabuckthorn due to increasing demand for its products, economic benefits, and multipurpose usage. Seabuckthorn cultivation began in Siberia during the 1930s (Kalinina & Panteleyeva, 1987) and has since been widely adopted worldwide, covering millions of hectares. Cultivars were developed for commercial cultivation with high yield, excellent berry quality, and adaptation

to diverse environmental conditions. Most seabuckthorn breeding programs aim to improve yield, fruit size, thornlessness, oil content, ripening time, frost hardiness, resistance to dried-shrink disease, fruit flavor, pollen, and ornamental value (Stobdan et al., 2022).

Cultivars for Commercial Cultivation

The wildy grown seabuckthorn plants cannot be grown commercially. High morphological diversity results in low fruit yield, non-uniformity in fruit ripening, differences in plant growth characteristics, and adaption to local climatic conditions. An ideal cultivar with specific fruit characteristics, maturity time, and plant growth characteristics for a particular region or cultivation technology are developed for commercial cultivation. Plant breeders often target yield, fruit size, thornlessness, harvesting, oil content, ripening time, frost hardiness, resistance to disease, fruit flavor, pollen, ornamental value, and good adaptability to diverse environmental conditions (Stobdan et al., 2022).

The need for breeding programs that generate numerous cultivars resulted from the need for plant materials that are well-adapted to local climatic conditions. Cultivated varieties from China are fast-growing and adaptable. However, they bear small fruits and more thorns, are difficult to harvest due to short fruit stalk, have a lower content of bioactive

substances, and are more prone to dried-shrink disease. In contrast, the cultivated varieties from Mongolia and Russia showed many promising agronomic traits, such as big fruits, few or no thorns, long fruit stalk, and high content of bioactive substances, but showed weak adaptability and are slow-growing (Ruan et al., 2013). By utilizing the hybridization method, desired traits from various locations have been combined to create new cultivars.

Desirable Characteristics of a Superior Cultivar

Fruit yield: Fruit yield depends on fruit size, fruit density on branches, and number of branches. Some cultivars with high fruit yield include Novost Altaya and Dar Katuni, developed by Lisavenko Institute. The cultivar Novost Altaya produces 11.3 tons, while Dar Katuni yields 12.7 tons per hectare at a spacing of 4 m × 2 m. The cultivar Chuyskaya yields 28.7 tons at 4 m × 2 m and 33 tons per hectare at 4 m × 1 m spacing (Singh & Zubarev, 2014).

Fruit Size: Large fruit is easy to harvest. Some of the cultivars with large fruit sizes include Elizaveta (1 g per fruit), Agurnaya (1.1 g per fruit), and Avgustina (1.1-1.4 g per fruit) (Singh & Zubarev, 2014).

Thornless: Developing thornless seabuckthorn has been one of the breeders' prime objectives. Berries are harvested manually, and the presence of the thorns hinders harvesting. Some thornless varieties are

Novost Altaya, Vitaminaya, Velikan, Prevoskhodnaya, and Alei (Kalinina, 1998). The thornless Mongolian and Russian varieties are being used to introduce thornlessness in fast-growing Chinese varieties with strong adaptability (Ruan & Li, 2005).

Harvesting: The peduncle is the stalk that joins the fruit to the stem, and its length determine the efficiency of mechanical harvesting. Fruit with a long peduncle is desirable for selection. Some varieties with long peduncle (6-10 mm) are Orangevaya, Luchezamaya, Chiulishmanka, and Ulala (Panteleeva, 1998). Fruit skin toughness is also an important trait for fruit harvesting. Skin damage during harvesting invites the growth of microbes and affects the quality of raw materials. Chuyskaya, Inya, and Avgustina are suitable cultivars for handpicking fruits. Individuals can collect 100-150 kg of fruits in 8 hours (Singh & Zubarev, 2014).

Oil Content: Seabuckthorn seed and fruit pulp contain oil that has applications in medicine, cosmetics, and nutraceutical supplements. Generally, varieties with high oil content have small berries, are difficult to harvest, and are only moderately productive. In contrast, varieties with moderate fruit oil content (5-8%) are high-yielding (Ruan et al., 2007). The cultivars Chechek and Dgemovaya contain 7.8 and 8% oil respectively (Singh & Zubarev, 2014).

Sweet fruit: Efforts have been made to breed sweet forms of seabuckthorn berries. The sweet fruits contain more than 9% sugar. Important forms are Tenga, 45-15-13, and 91-81-3 (Zubarev, 1998).

Resistance to insect pests and diseases: Only a few insect pests and diseases are reported in seabuckthorn. Dried-shrink disease and carpenter moth (*Holcocerus hippophaecolus*) are serious problems. It is desirable that the commercial varieties be resistant to various insect pests and diseases.

Extended fruit ripening: Seabuckthorn berries are delicate and must be harvested soon after ripening. To harvest berries for an extended period, early, mid, and late fruit ripening cultivars are desirable. Cultivars that flower later and ripen earlier are preferred in high-altitude regions. Late blossoming is an important factor in protecting from damage caused by spring frost and, therefore, a desirable trait in regions experiencing spring frost.

Winter hardiness: Winter-hardy cultivars are required in most temperate zones experiencing severe cold winters. Harsh and unpredictable winters threaten the plant's survival (Tang & Tigerstedt, 2001).

Ornamental value: Seabuckthorn berries vary in shape and color. Typically, the berries are globose to egg-shaped, and their color ranges from yellow to red. The combination of fruit shape and size, together with

the contrast between the color of the fruit and leaves, contributes to the ornamental value of this plant (Yao & Tigerstedt, 1994). The cultivar 'Friedorfer Orange' was developed at the Experimental Station of Floriculture and Nursery Management, Germany, for ornamental value.

Elite Cultivars

Most known cultivars have been developed in Russia, Canada, Germany, and China. Seabuckthorn has received little attention for breeding high-yielding varieties in India (Stobdan et al., 2022). Some elite seabuckthorn cultivars and their characteristics are:

Avgustina: It is a winter-hardy Russian cultivar with large fruit (1.1-1.4 g per fruit). The long peduncle (5-6 mm) makes harvesting the berries easy. The fruit is suitable for fresh consumption and processing (Singh & Zubarev, 2014).

Elizaveta: A winter hardy Russian cultivar bearing large fruit (0.8-1.1 g per fruit) with a long peduncle (5-6 mm). The fruit has a pleasant aroma and is suitable for fresh consumption and processing (Singh & Zubarev, 2014).

Chuskaya: It is the most widely cultivated in Russia and many countries. The fruit is sweet and easy to harvest by handpicking (Singh & Zubarev, 2014).

Chechek: Chechek is a winter-hardy Russian cultivar. The fruits are large (1.1-1.4 g per fruit) and easy to harvest with long peduncles (5-6 mm). They are suitable for fresh consumption and processing (Singh & Zubarev, 2014).

Sibirski Rumyanets: It is a winter-hardy, early-ripening Russian cultivar. It is resistant to drought, and the fruits have a sour taste and unique fragrance (Belykh, 2003).

Capriz: Capriz is an early-ripening Russian cultivar. The large fruits have a sweet taste and pleasant fragrance and are suitable for fresh, processing, and freezing (Belykh, 2003).

Frugana: It is an early ripening cultivar developed in Germany. It bears medium-sized fruits (0.4-1.0 g per fruit) that are easy to harvest. The fruit juice has a mild taste (Albrecht, 2003).

Harvest Moon: It is a cultivar developed in Canada. It is a small, compact bush-type plant with a 2.8-m height. Thorns are mostly absent. The fruit is large (8.7 g per fruit) with sour-sweet. The pedicels are long (4.1 mm), hence easy to harvest (Schroeder et al., 2014).

Indian-Summer: It is a hardy and above-average drought-tolerant cultivar. Each plant yields 4-5 kg of fruit. The cultivar is well adapted to Canadian prairies (Li, 1999).

Qiuyang: Qiuyang is a cultivar developed in China. The plant is hardy and highly drought-resistant. The fruits ripe early and are large-sized (0.7 g per fruit) (Zhang et al., 2010).

Liuyeshaji: The cultivar was selected from a wild plant growing in Tibet. It is a thornless cultivar with no or few thorns. Each plant reaches a height of 3-5 m (Li et al., 2009).

Soil

Seabuckthorn grows well in various soil types, including sandy, rocky, saline, and ravine soils. Light mineral-rich soil with a pH near neutral (pH 6.5-7.5) is ideal. Best growth occurs in deep, well-drained, sandy loam with ample organic matter.

Propagation

Seabuckthorn plants can be propagated by seed, but there are better approaches than this method for commercial cultivation. The resulting plants from the seed are not genetically identical to the parent plant. The seedlings display variable growth form, height, and fruiting. Additionally, the plants grown from seed can either be male or female, and more females are desired for commercial purposes than males. Besides, the female plants obtained from the seed started bearing fruits in 4-5 years, whereas the plant propagated from cuttings set fruit in 2-3 years.

Vegetative propagation methods produce offspring that are genetically identical to the stock plant. Hardwood cuttings are commonly used in propagating deciduous woody plants as they are an easy and inexpensive method. Rooting can be done in nurseries without the need for any special equipment. Vegetative propagation through pencil-thickness hardwood cutting is the method of choice for the propagation of the species.

Cuttings: Standard methods have been established for the propagation of hardwood stem cuttings that are approximately the thickness of a pencil (7.5 mm diameter). Seabuckthorn stem cuttings of 14-16 cm in length are taken from male/female plants in the first week of April (in cold, arid Ladakh conditions). The cuttings are first soaked in tap water for 24 hours, then treated with 500 ppm indole-3-butyric acid (IBA) for 1 minute after removing the thorns and cutting the base (prepare IBA solution in 50% isopropanol). The cuttings are then planted in the nursery bed at a 6"×6" gap between cutting to cutting and row to row. During field preparation, 3.5 kg of farm yard manure is applied in each plot (4.6 m²). Watering the plants once a week is necessary. The use of black plastic mulching resulted in 10% higher rooting success. Studies conducted in Ladakh showed that cutting thickness significantly affected rooting success. The highest rooting percentage is observed in pencil-thickness cuttings (7.5 mm diameter), followed by cuttings with less than half the pencil thickness (3 mm diameter) and

those with more than pencil thickness diameter (11 mm) (Dolkar et al., 2016).

Suckers: Seabuckthorn has a dense suckering habit. Each plant produces 13-65 suckers that can be easily separated from the parent (Dwivedi et al., 2006). The individual suckers have been utilized as planting material to maintain the true-to-type consistency.

Seed: Seabuckthorn can also be propagated through the use of seeds. Between September and October is the ideal time to collect seeds from fully matured fruits. The seeds exhibit short-term physiological dormancy, which means they should not be sown right after harvest. Seabuckthorn seeds can be stored at room temperature for up to 6 years without significant loss of viability in cold, arid Ladakh conditions (Korekar et al., 2013). However, in most places, dry seeds can be kept satisfactorily at room temperature for 1-2 years (Slabaugh, 1974). It is recommended to soak the seed in water for 48 hours before sowing. Seeds are sown in spring when the soil temperature at 5-10 cm depth rises to 15-20°C (Rongsen, 1992). Sowing should be done at 1-2 cm soil depth.

Light

Seabuckthorn can only grow in well-lit and unshaded areas. Throughout all stages of development, the plant cannot tolerate shade. Seedlings or cuttings die if

shaded. Mature seabuckthorn produces smaller crowns and has lower yields if shaded.

Orchard Establishment

A single or double hedge row system for seabuckthorn plantation in an orchard is recommended. In a single hedge row system, the spacing between row to row is maintained at 2-4 m for pure cultivation and at 4-5 m for intercropping (Dwivedi et al., 2006). For proper management, the plants should be placed in rows with a 1-2 m distance. Spacing is an essential factor in determining fruit yield.



A newly established seabuckthorn orchard in Ladakh

The method of plantation and spacing needs to be modified based on the cultivar, soil fertility status, and climatic conditions to obtain good fruit yield. For

economic reasons, the ratio of male to female plants is essential, as the number of female trees directly affects the total yield. Recommended male plants in an orchard vary from 6-12%. An uneven distribution of male and female plants may occur if unknown seedlings are planted. Weed control is crucial for promoting the growth of newly planted seabuckthorn seedlings. Intercropping of gladiolus with seabuckthorn has been successful (Stobdan et al., 2011a).

Transplanting

Spring is the best time for planting seabuckthorn when it is in its dormant stage. The plants should be watered once every week after transplanting. In orchard planting, spacing 1-1.5 m within the row and 3-4 m between rows is recommended. Rows should be oriented north-south to provide maximum light.

Irrigation

Seabuckthorn can tolerate drought but is a moisture-sensitive plant, especially in spring at flowering and fruit development stages. For economic reasons, seabuckthorn orchard establishment should be restricted to areas receiving a minimum of 400 mm annual precipitation unless irrigated. The optimum soil moisture content for mature plants is around 70%. Mulching should be practiced to reduce the loss of soil moisture. Drip irrigation can be encouraged for

optimal growth of plants and improved water use efficiency. The use of black plastic during the first 3-4 years has been found to deal with moisture conservation and weed control. It is susceptible to water logging conditions; thus, water stagnation should be avoided at all stages.

Manure and Fertilizer

Seabuckthorn plants require little nitrogen; however, they respond well to phosphorus fertilizer. Well-rotten Farm yard Manures (FYM) @ 45 tons per hectare should be used, depending upon soil conditions (Li & McLoughlin, 1997).

Pruning

Pruning is essential to train branches, promote growth, and facilitate harvesting. Moderate pruning increases the yield and fruiting life of the plants. The pruning process involves the removal of overlapping branches and cutting long branches to encourage lateral shoot development. Mature fruiting plants should be pruned to allow more light penetration.

Mulching

The use of black plastic, particularly in cold mountain regions during the first 3-4 years, promotes quick growth of the plant. It increases soil temperature by 1.6-3.6°C. Higher soil temperature, particularly in the

plant's early growth, is beneficial in cold climatic regions such as the trans-Himalayan Ladakh region. Further, it reduces the cost of weed control and keeps the soil moist.

Yield

Fruit yield depends on species, cultivar, plantation age, and cultural practices. Some cultivars with high fruit yields include Novost Altaya and Dar Katuni, which Lisavenko Institute developed. Novosti Altaya produces 11.3 tons, while Dar Katuni yields 12.7 tons per hectare at 4 m × 2 m spacing. Cultivar Chuyskaya yields 28.7 tons at 4 m × 2 m and 33 tons per hectare at 4 m × 1 m spacing (Singh & Zubarev, 2014). Under Ladakh conditions, where the plants grow naturally without any management practices, the fruit yield varies from 0.2-8 kg per plant. Leaf yield also depends on several factors. The leaf yield of 5-year-old nine Russian varieties ranges from 1.0 to 2.4 kg per tree (Morozov, 2007).

Insect-Pests and Diseases

Very few insects, pests, and diseases have been reported in seabuckthorn. Green aphids (*Capithophorus hippophae*) are reported to be the most damaging insect. These aphids cover the lower side of the leaves and suck the cell sap. Thereby, leaves become yellow and die. Aphids can be controlled with insecticide or by biocontrol agents.

Death Hawk Moth (*Acherontia styx*) is a prominent pest and does considerable damage to the seabuckthorn plant at the larval stage in Lahaul Spiti in Himachal, India (Bhagat et al., 2003). Defoliating beetles belonging to sub-family Melolonthinae viz. *Brahmina cariaceae* (Hope) and *Brahmina* spp. also attacked seabuckthorn. Although these beetles' attacks are sporadic, they reduce plant vigor and fruit-bearing. It has been observed that the Indian Meal Moth (*Plodia interpunctella* Huber) causes damage to berries in storage. Fruit fly (*Phagoletis batava obscuriosa* Kol), defoliator brown tail moth (*Euproctis chrysorrhoea*), clear wing moth (*Synathedon hippophae*), root borers (*Cossus* spp and *Holotricha* sp), Psyllido (*Cacopsylla* spp), trunk borer (*Asias haladewdri*), *Gelechina hippophaella*, *Hippophaetrizo chuensis* and *Hippophaetrizo guauguwuni* have also been reported on seabuckthorn from various regions of the world. Root rot caused by *Rhizoctonia solani* Kuhn is reported from Uttarakhand, India (Singh et al, 2007). Mice and rats are other pests that can destroy and girdle the trunk or chew up roots.

Few diseases in seabuckthorn have been reported, and the most serious one is scab, which strikes female and male plants, especially in a rainy spring-summer period (Dolgacheva & Aksenova, 2003). Scabs appear as dark-brown tuberous spots, which gradually become black and glistening. The spots appear either on the leaves or the fruits.

Dried-shrink disease is a fungal disease that destroys seabuckthorn and halts commercial production. In China alone, the disease decreases fruit yield by 30–40%, eliminating 4,000 ha of mature plantations annually. The symptoms include yellowing of plant tissues; as the disease develops, the bark putresces, dries, and shrinks, and finally, the plant dies. The pathogen infects older plants (above three years), often at the plant base. Fruits do not develop properly and drop early in June. The branches bend down and dry up, usually before July (Ruan et al., 2010).

Fusarium wilt is also reported in seabuckthorn, which is marked by tarnishing and falling leaves. The fruits prematurely become brown and wither but don't fall. Cutting off and burning the damaged branches is an efficient method of plant protection.

Verticillium wilt also appears in the middle of summer. The leaves become yellow and fall. The fruits prematurely dry up, and the plant dries out very quickly. The shoots become black-cherry colored with the swelling of the bark. The diseased plant should be uprooted and burnt.

Stumping in Wild/Forest Plants

Seabuckthorn plants, when grown in areas with forests or barren land, tend to experience a decrease in growth and productivity after approximately ten years (Tian et al., 2021). To address this issue, it is

recommended to cut the aboveground portion of the plant to a height of 15-20 cm. This practice helps to stimulate fine root growth, prevent aging, and speed up the renewal and rejuvenation process of the plants (Wang et al., 2023).

Gender Difference in Plant Adaptation

In species with separate sexes, both male and female individuals adapt differently to environmental stress in terms of their physical characteristics and biological processes. Females typically invest more resources into reproduction, resulting in a greater reproductive effort. As a result, females tend to prioritize fruit development over vegetative growth or protection. This significant investment in reproduction can lead to oxidative stress and cellular injuries, especially under adverse conditions. Consequently, there are gender differences in how plants adapt to environmental stress and in the nutrient contents of their vegetative parts.

Freeze tolerance responses: Seabuckthorn is known for its remarkable ability to tolerate various environmental stresses. Male plants have a better self-protection mechanism than females when coping with cold and freezing conditions. On the other hand, female plants prioritize their resources in fruit development rather than focusing on protection. When planting seabuckthorn in cold mountainous regions, it is vital to provide extra protective measures

to females to ensure higher survival rates (Li et al., 2005; Dolkar et al., 2017a).

Male-biased population in a stressful environment:

When resources become scarce, the environment tends to favor males and stressful environments negatively impact females (Li et al., 2003). Besides, male plants are more adaptable to changing climates, which may lead to a male-biased population in the event of climate change (Dolkar et al., 2019).

The development of seabuckthorn-based products has led to tremendous commercial impact, with demand exceeding supply. Thus, cultivating seabuckthorn has become a compelling solution, with cultivation packages already in place in several countries. However, it is crucial to consider the local conditions carefully before selecting the best practices. Male plants have a more effective self-protection mechanism that helps them cope with cold and freezing conditions. Therefore, providing extra protective measures to female plants is crucial to ensure their higher survival rate.

Chapter-9

Harvesting Berries and Leaves



Manual harvesting of berries using the 'beat the bush' method

Seabuckthorn berries and leaves, which have economic value and medicinal properties, are collected. Seabuckthorn berries ripen in August in Russia and European countries. However, they ripen in September in the mountain regions of India and China. Due to the thin fruit skin, about 15% of the berries

burst during harvesting (Singh et al., 2008). Overripe berries and leaves are challenging to harvest.

Fruit Harvesting

Seabuckthorn berries ripen in late August to mid-September, and harvesting starts soon after ripening. In some countries, the berries are harvested during winter when the fruit is frozen. Berry harvesting typically occurs over a brief period of 20-30 days. The berries are small, soft, delicate, and juicy; and thus susceptible to spoilage (Li & Schroeder, 1996). Therefore, the fruits must be transported quickly to a processing facility and immediately cooled to temperatures around 6°C or less to slow the growth of microorganisms.

Berry harvesting is laborious and time-consuming, contributing to approximately 58% of the total production cost (Li & Schroeder, 1996). Harvesting the small and delicate berries is difficult due to their short pedicel, lack of abscission layer, and presence of thorns. Harvesting fruit is comparatively easier in the early morning hours. Different methods are used to collect the berries, including direct and indirect harvesting techniques.

Beat the bush: Seabuckthorn berries are manually harvested in September using the 'beat the bush' method. A plastic sheet is placed on the ground under the canopy. The tip of a branch is held with one hand

and beaten with a stick with the other. The berries that fall on the plastic sheet are collected in a bucket and transported to the processing unit in a 10-kg capacity food-grade plastic tray. Harvesting during the early morning hours before sunrise causes less damage to the plant, and it is easier to harvest the fruit. This method is being practiced extensively in Ladakh, India (Stobdan et al., 2011).

Hand-picking: In this method, fruits are picked by hand, either from the ground or from the plant. The method is labour-intensive and time-consuming. An individual can harvest 5-10 kg of berries in 8 hours (Gupta & Singh, 2003).

Branch Cutting: Branches bearing fruit are pruned, and berries are harvested manually or mechanically. Quick-freezing the branches makes berry harvesting easier by vibration. Harvesting by this method causes a significant drop in fruit yield the following year.

Jaw-tooth and Brush Harvester: The device consists of two jaws with teeth and brushes. The jaws are closed onto a fruiting branch, and the fruits are rasped off to fall through a fabric chute into a bag-type container. The device weighs 600 g, and an individual can harvest 40-100 kg of fruit in 8 hours (Gupta & Singh, 2003).

Fruit Comb: A simple method used in Mongolia. A steel wire hook is used to harvest berries by combing the

fruits in two directions. An individual can harvest 40-100 kg of fruits in 8 hours (Gupta & Singh, 2003).

Hormone Treatment: Using 2,000 ppm ethephon spray on the seabuckthorn plant reduces the fruit detachment force by 30%, thereby reducing the harvest time (Trushechkin et al., 1973).

Air Suction: This is an indirect method of berry harvesting in which there is no direct contact with the fruit. Berry is harvested by force air suction.

Mechanical Shaker: Trunk and branch mechanical shakers are used in several countries for berry harvesting to reduce the harvesting cost. However, two major problems have been encountered with such harvesters. The bark gets damaged, and the vibration force required for berry harvesting varies depending on the cultivars.

Portable Vibration Picker: A portable mechanical vibration seabuckthorn berry picker was developed in China. It harvests 80% of the berry with an efficiency of 48.9 kg per hour (Fu et al., 2014).

Leaf Harvesting

Seabuckthorn leaves are narrow, elliptical, 20-40 mm in length, 2-4 mm wide, and silvery. They are usually collected in October and November. Leaf harvesting has been a major problem. Harvesting by hand is both

time-consuming and labour-intensive. Efforts have been made to develop mechanical leaf harvesters in Canada, where manual labour is either unavailable or economically prohibitive (Mann et al., 2003).

Manual Harvesting: A plastic sheet is placed on the ground under the canopy. The tip of a branch is held with one hand and beaten with a stick with the other. The leaves that fall on the plastic sheet are collected and shade-dried. The method is widely practiced in Ladakh from October to November (Stobdan et al., 2011).

Hand-held Prototype: A hand-held prototype consisting of a pair of hydraulically powered, rotating brush heads at the end of a tubular aluminium frame and other accessories has been developed. The best harvest rate observed was 10.5 g per sec using 1.5 mm diameter nylon bristles rotating at 525 rotations per minute (rpm). However, several modifications are needed for practical application (Mann et al., 2003).

Trailer-mounted Prototype: The prototype has been developed to overcome the problem of the excessive weight of the hand-held prototype. The harvest rate ranged from 3.8-17.0 g per sec using 1 mm diameter polypropylene bristles rotating at 500 rpm. However, for practical application, several modifications are needed (Mann et al., 2003).

Harvesting berries and leaves is laborious and time-consuming. Several harvesters have been developed to make the task easy and economical. However, the principal source of the berries has been areas planted with seabuckthorn for land reclamation and erosion control. Therefore, mechanized harvesters find no use. In India, seabuckthorn is harvested from the wild, rendering most developed harvesting equipment unsuitable. 'Beat the bush' method is being practiced extensively for berry harvesting in Ladakh, India.

Chapter-10

Postharvest Handling of Berries



Freshly harvested seabuckthorn berries with leaves and other debris

Seabuckthorn fruits are small, soft, delicate, and juicy. Since they are highly perishable, long-distance transportation of fresh berries is difficult. Fresh seabuckthorn berries are processed on-site, separating the pulp, seed, and hull for further value addition.

Berry collection is best done in the morning to avoid damaging the shrub while harvesting. Seabuckthorn

berries, when overripe, carry a strong musky odor with a rancid taste, which can be detectable even in the fields. Therefore, berries must be harvested at the correct stage, quickly transported to the processing plant, and should be cooled immediately to 4-6°C to retard spoilage. If the berries are to be stored for over a few days, they should be frozen, preferably by quick freezing techniques (Li, 2002). Fruit can be frozen to -18°C for long-term storage without further loss of ingredients. The pulp can be stored using preservatives such as potassium metabisulfite and benzoic acid.

Berry Cleaning

Berry inspection and cleaning are essential to remove damaged berries, twigs, leaves, and other debris collected during harvesting. The first step in cleaning berries is to roll them on a wet surface (L×W: 7-8 feet × 3 feet) placed at an angle of 40-50°. The wet surface traps leaves and damaged berries, which are manually removed. Twigs and other debris not sticking to the wet surface are removed by hand. The berries are washed in water and sieved to remove leaves and damaged berries. Water is drained from the berry before juice extraction (Stobdan et al., 2015).

Juice Extraction and Storage

Berries are typically processed for juice extraction using a machine called a pulper. When cleaned berries are fed into the pulper, pulp (74%) and cake (24%)

containing unbroken seeds are obtained. The freshly extracted juice is collected in food-grade drums with a 50 kg capacity. Potassium metabisulfite is added as a preservative. The pulp is the main marketable component. Occasionally, the pulp is passed through a filtration unit to get cake and a clear aqueous solution (pH 2.7; TSS 10.6°Brix). The cake with unbroken seeds is sun-dried, and the seeds are separated from the hull by rubbing the dried residue. Both the seeds and hulls contain valuable oil, which can be sold separately or with the pulp (Stobdan et al., 2015). The seed is in great demand as it contains omega-rich oil, which is generally extracted by supercritical CO₂ extraction methods.

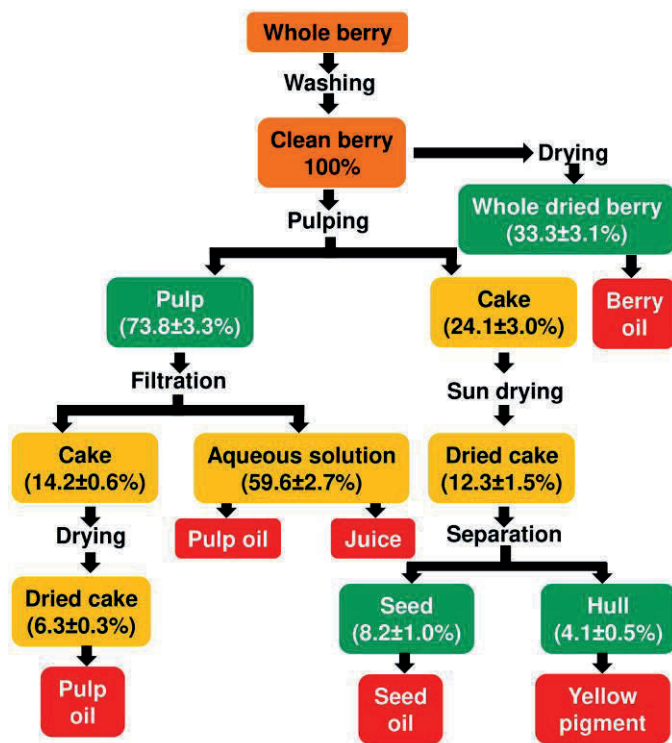
Juice from ripe fruits is extracted by pressing or centrifugal techniques and can be stored under refrigeration for a few days. It requires pasteurization or freezing for long-term storage. Aseptic filling at low temperatures improves the storage of seabuckthorn juice. If fresh juice is allowed to stand for one or two days, it separates into three phases: an upper floating particulate phase, a central liquid portion, and sinking particulate sediment. This separation is undesirable from a consumer point of view (Kleinschmidt et al., 1996). If the pulp oil is not removed, it forms an oil layer on the juice surface. The centrifugal reduction of oil below 0.1% eliminates the floating problem (Beveridge et al., 1999). For preservation purposes, sterilizing/pasteurizing the juice is necessary. High-temperature-short-time (HTST) processes of 80-90°C

for several seconds are preferred (Liu & Liu, 1989). Storing juice for more than 6 months can cause it to brown, but refrigerating it at 4°C can extend its shelf life.

During processing and storage, some changes occur in the bioactive compounds. Phylloquinone (vitamin K) loss during industrial juice production has been reported at about 36-54% (Gutzeit et al., 2007). Similarly, the total ascorbic acid loss during industrial juice production is about 5-11%. The production of concentrated juice results in 50% depletion of ascorbic acid (Gutzeit et al., 2008). Storage of seabuckthorn juice for seven days at ambient temperature (25°C) results in 18% degradation of pantothenic acid. Therefore, the biochemical content of the juice depends on the processing technologies and the equipment used. Storage of seabuckthorn concentrated juice results in a gradual increase in total soluble solids, titratable acidity, reducing sugars, and total sugars, while ascorbic acid content and consumer acceptance decrease during 120-day storage trials (Ali et al., 2011).

A schematic diagram of the processing of fresh seabuckthorn berries is shown in the flow diagram. During the cleaning process, a loss of 4.7% (w/w) during rolling and 8.5% (w/w) during washing is recorded, which together amounts to 13.2% (w/w). Large quantities of damaged berries, leaves, twigs, and other debris in the harvested berries are primarily due

to the crude method of berry harvesting (Stobdan et al., 2015). There is a need to develop mechanical harvesters, and seabuckthorn planting must be done systematically to enable easy berry harvesting.



Primary processing of seabuckthorn berries. Values in the bracket show the percentage (w/w) of various components obtained from processing 100 kg of clean berries. Products shown in green boxes are typically sold from Ladakh for further value addition outside of the region (adapted from Stobdan et al., 2015).

Sun Drying

Sun drying the whole berry is also a common practice. However, the drying method is tedious and generally takes over a month to obtain the dried product. Besides, it requires a large drying area and protection from rain and wind. Therefore, open sun drying is a cumbersome method involving risk. Regionally suited low-cost electric dryers need to be used. On average, 33.3 kg of whole dried berries is obtained when 100 kg of cleaned berries is sun-dried.



Previous year's dried berry on the shrub

Oil Extraction

Seabuckthorn contains pulp and seed oil. Pulp oil can be separated by centrifugal force or using organic solvents such as hexane. Since seabuckthorn oil is

mostly used in the cosmetic industry or as a nutraceutical supplement, newer techniques, such as supercritical carbon dioxide extraction under high pressure, are preferred. Drying of pulp and seeds provides extended shelf life. Air-dried and freeze-dried seeds yield similar extraction yields to hexane, whereas air-dried pulps yield better than freeze-dried pulps.

Pigment Extraction

The yellow pigment can be extracted from seabuckthorn waste after juice extraction with a low alcohol concentration after concentrating the suspension to 11 to 13° Brix (Chen et al., 1995; Liu et al., 1989). The pigment contains flavones, carotene, and vitamin E. Pressure has the most significant influence on extraction, with yield increasing with extraction pressure.

Precautions

- a) Seabuckthorn is a highly thorny plant. Workers harvesting the berries and leaves are at risk for eye injuries. It is crucial to be mindful and take necessary precautions to protect the eyes to avoid eye injuries that may lead to vision loss. Protective clothing and safety glasses are recommended.

- b) The berry must be harvested at the correct stage. Underripe berries yield less juice, while overripe berries carry a strong, musky odor with a rancid taste.
- c) Fruit should be harvested in the early morning. Harvesting is comparatively easier in the morning hours and causes less damage to the fruit and the plant.
- d) Seabuckthorn fruits are highly perishable and cannot be transported long distances. The berry must be transported quickly to the processing plant and processed immediately. If the berries are to be stored for more than a few days, they should be frozen, preferably using quick freezing techniques.
- e) Food-grade plastic sheets/trays/drums should be used during the entire process of berry harvesting, processing, and storage.
- f) For juice extraction, the berries should be free from damaged fruits, twigs, leaves, and other debris that gets collected during harvesting.
- g) During drying, precautions should be taken so that children and domestic cattle do not roam around the drying yard.

Seabuckthorn: The Superfruit

Seabuckthorn berries are small and delicate. They spoil quickly if not processed on the same day. Freshly harvested berries are unsuitable for long-distance transportation; hence, processing is necessary near the harvesting area. After primary processing, the pulp, seed, and hull are used to add further value.

Chapter-11

Traditional Uses



A pile of seabuckthorn wood for firewood purpose at Shey palace, Ladakh

Natural resources are assets for development. Many of the world's wealthiest nations have achieved affluence and technological advancements by exploiting their natural resources. Traditionally, when resources are scarce, people build mastery in judicious use of their available resources. Mountain people have mastered utilizing seabuckthorn for various uses, primarily governed by ethnic preference, agroclimatic conditions, sociocultural ethos, and religion. Recent

studies have confirmed and expanded the traditional uses of seabuckthorn, and various products have been developed for their nutraceutical and medicinal properties.

Traditional Medicine

Seabuckthorn is a medicinal plant used in traditional Tibetan, Indian, Chinese, Mongolian, and Tajikistan medicine. It has been valued for its medicinal properties since the 8th century, as documented in the Tibetan medicinal classic *rGyud Bzi* (Four Medical Tantras). There are around 300 popular seabuckthorn-based formulations in various pharmacopeias of *Sowa Rigpa* (Tibetan medicine). In Ladakh region, even today, *Amchi* (local traditional doctors) often prescribe preparations from seabuckthorn for treatment of common problems like indigestion, throat infection, gynaecological problems, ulcers, gastritis, bronchitis, acidity, diarrhea, hypertension, blood disorder, fever, tumour, gallstone, cough, cold, food poisoning, and many more. In Turkey, seabuckthorn has been traditionally used as an antitussive and aphrodisiac (Cakilcioglu et al., 2011). Modern-day laboratory and clinical studies confirm the efficacy of seabuckthorn for its medicinal properties for the treatment of oral mucositis, vaginal mucositis, cervical erosion, duodenal ulcers, skin ulcers, cancer, sluggish digestion, stomach malfunctioning, neoplasia, thrombosis, hepatic injury, tendon, and ligament injuries, and many more.

Wine

During Genghis Khan's reign, Mongolians pioneered preparing seabuckthorn wine using highly acidic berries. The wine was exclusively served to royal families and nobles (Chen et al., 2023).

Jam and Juices

There is a tradition of making jam and juice from seabuckthorn fruit on the Finnish coast along the Gulf of Bothnia in Finland (Rongsen, 1992).

Soil Fertility Improvement

Growing seabuckthorn for fencing around agricultural fields is a beneficial practice. Farmers in Nubra Valley, Ladakh, mix soil from densely growing seabuckthorn areas with low fertile fields to enhance soil fertility (Stobdan et al., 2013). Studies show that seabuckthorn's root nodules have a symbiotic relationship with a nitrogen-fixing bacterium called *Frankia*. Seabuckthorn fixes 180 kg of nitrogen per hectare per year, improving soil fertility (Jike & Xiaoming, 1992).

Firewood

Cold deserts are characterized by sparse vegetation and severe cold winters. This requires the cultivation of fast-growing plant species that can be used as firewood for cooking and heating homes.

Seabuckthorn has proven to be a popular green energy plant in the region since the stem, branch, and root are being used as firewood. The calorific value of dry seabuckthorn is 4,785 calories per kg. The shrub is fast growing and can be stumped after every 3-5 years. Since the shrub grows fast and tolerates repeated cuttings, it reduces the harvesting pressure on other native woody plant species, such as poplar, willow, and juniper. A six-year seabuckthorn plantation on one hectare can produce 18 tons of firewood, equal to nearly 12.6 tons of standard coal (Rongsen, 1992). Firewood is popular, especially in monasteries and during ceremonies where high calorific firewood is required. Most monasteries in the Ladakh region maintain their seabuckthorn growing areas, popularly called *Tsokshing*, to ensure a regular supply of firewood. In the local dialect, *Tsok* means "forest", and *Shing* means "wood". This reinforces the widespread use of seabuckthorn as a regional firewood plant (Stobdan et al., 2013). The shrub is used as firewood to roast barley grains and prepare the staple food *Tsampa*. The use of seabuckthorn as firewood is slowly declining since alternate sources are available.

Fencing

Due to limited cultivable land, agricultural fields are most valued in cold deserts. The dense and thorny shrub is traditionally planted around agricultural fields and plantation sites to protect against stray animals and pedestrian movement. The thorny dried twigs and

branches are placed along the boundary walls of residential properties and fields. The traditional use of seabuckthorn for fencing is still widespread in Ladakh, but there is a slow decline since alternate fencing material is available at subsidized prices.



Use of dried seabuckthorn branches for fencing

In Zanskar Valley in Ladakh, dried seabuckthorn branches, stems, or whole plants are staked around the edge of the house's flat roof. It gives protection against heavy snowfall. The decayed biomass keeps the rooftop soil intact and protects against the melted snow water that otherwise percolates inside the house. Besides, the enclosure is a protective shelter for keeping the lambs and goat kids on the rooftop (Stobdan et al., 2013).

Tree Guard

Plantation of timber trees like willow and poplar is an essential activity in cold mountain regions. The cuttings during their initial years need the utmost care, especially from stray animals. Traditionally, to protect the plants from animals, cuttings are wrapped with thorny branches as tree guards.



Use of seabuckthorn as tree guard in Ladakh

This practice effectively increases the plants' survivability percentage during the early stages in the Ladakh region (Stobdan et al., 2013). A decline in the traditional use of seabuckthorn as a tree guard has been observed, with empty tinned food packaging materials now being used as an alternative.

Soil and Moisture Conservation

Cold deserts are known for their high wind velocity, which leads to environmental degradation. Seabuckthorn windbreaks are an effective means of preventing wind erosion in open areas. The shrub can resist the drying effect and physical injuries caused by the wind. Seabuckthorn develops an extensive root system. In 10-12-year-old shrubs, the horizontal root extends up to 537 cm and vertically up to 127 cm deep, thus making it an ideal plant for soil binding (Dwivedi et al., 2006). Due to the extensive root system, seabuckthorn plantation is done around the water channel to check erosion due to water flow. Seabuckthorn plantation intercepts precipitation by the lush crown layer, redistributing precipitation and controlling surface runoff. It improves soil physical characteristics and fertility, thereby improving soil water-holding capacity.

Temporary Bridge

It has been observed that the shrub is being used in a unique way in Ladakh. The seabuckthorn plant is cut above the ground and placed horizontally on two opposite sides of the river during winter. This supports floating ice blocks in forming a stable ice cover. With time, the ice cover from the two opposite sides joins to create a long, thick, single ice sheet connecting the two opposite river sides. The ice cover is used to cross the river when bridges are either far away or do not

exist. The art is not practiced anymore as it is unknown to most people (Stobdan et al., 2013).

Charcoal

Blacksmiths use high-calorie charcoal from burnt seabuckthorn branches and stems. When burned, seabuckthorn-based charcoal does not leave behind much ash compared to those obtained from other native woody plant species. It is the most preferred charcoal and is commonly used by blacksmiths in the Ladakh region (Stobdan et al., 2013).

Building Construction

Seabuckthorn stems, hard and durable, are often used as roofing material in place of native woody plant species like willow and poplar. However, due to the plant's shrub-like nature and its medium-length stem, it is not a popular roofing material. Besides, the stem does not grow straight and restricts its use to constructing only small rooms such as cow sheds and toilets. Seabuckthorn branches are often cut into small pieces and mixed with mud for erecting walls. The branches function as modern-day iron bars, reinforcing walls by keeping the mud intact. However, the method is slowly declining since modern-day building materials are readily available (Stobdan et al., 2013).

Religious Rites

Seabuckthorn stem is essential for performing specific religious rites in Tibetan Buddhism. It is used as a supporting stand for preparing *Chotpa*, which is a ritual cake. In some pockets of the Ladakh region, it is popularly believed that placing seabuckthorn near the main entrance door wards off evil spirits (Stobdan et al., 2013).



Use of seabuckthorn for religious rites in Ladakh

Agricultural Implements

Seabuckthorn stems are hard and often used as a handle for agricultural implements such as sickles, axes, saws, etc. The wood is smooth and free from thorns. However, its use is slowly declining since the

wood is heavy, and alternate lighter materials are readily available (Stobdan et al., 2013).

Fabric Dye

The tribes in Sikkim and Arunachal use seabuckthorn juice to colour wool fabrics (Basistha, 2009; Vairale et al., 2020).

Metal Polishing

In Bhutan, the fruit of *H. salicifolia* is used for polishing gold and silver (Rongsen, 1992). In Ladakh, the fruit juice is used to polish metal surfaces.

Nutritional Supplement

Attributes of seabuckthorn to meet nutritional requirements in high altitudes are well recognized. Seabuckthorn berries are highly acidic and not generally consumed directly. Children often consume ripe berries from selected plants that bear relatively sweeter fruit. Seabuckthorn leaves are often brewed into tea due to their high antioxidant content. However, the use of seabuckthorn leaf as tea is declining since it does not have an attractive color, and alternate sources are readily available in the market (Stobdan et al., 2013).

Traditional Uses

When resources are scarce, people build mastery in judicious use of their available resources. People in the mountains have mastered utilizing seabuckthorn for various uses. Some of the traditional uses are declining, while the uses related to nutraceutical and medicinal properties are receiving increasing attention. There are around 300 seabuckthorn-based formulations in Tibetan medicine, and researchers are trying to validate and use them to treat various diseases.

Chapter-12

Seabuckthorn for Sustainable Development of Cold Desert

The cold desert region is characterized by extremely low temperatures and low precipitation, a thin atmosphere with high UV radiation, high wind velocity, and sparse plant density. In India, the cold deserts spread across 94,809 km². The majority of this area (87.4%) is located in the Union Territory of Ladakh, while the remaining 12.6% is in the Lahaul & Spiti region of the State of Himachal Pradesh (Sharma, 2002). Because of harsh environmental conditions, the cold desert region is sparsely populated.

The economy of the cold desert region of Ladakh has undergone significant changes in the last few decades. The population of Ladakh has increased from 105,292 in 1971 to 274,289 in 2011. Besides the military personnel deployed in the region, Ladakh's floating population is also growing rapidly. The number of tourists visiting Ladakh has increased significantly in recent years, and tourism is emerging as a primary industry generating revenue in the region. However, the benefits of tourism are unevenly distributed

among the population in Ladakh, leading to income inequality in the area. Investment in developing a land-based economy is necessary to enable farmers to continue working on farmland.

Seabuckthorn as a Means for Sustainable Development

The Himalayan region hosts many unique species, which offers significant opportunities for area-specific comparative advantages. However, the local resources with the potential for comparative advantages are often ignored, and the path towards development continues to be searched through conventional methods. Despite being once considered a problematic plant, seabuckthorn is now being recognized as a tool for sustainable development in the Himalayan region (Stobdan et al., 2017). The production of seabuckthorn-derived products has led to commercial opportunities to the extent that the demand for the berry from the area now exceeds the supply. With the growing demand for seabuckthorn, the berry harvest from current resources is expected to increase to 1,509 metric tons in 2030, compared to 985 metric tons in 2021 (Bindal, 2016).

Seabuckthorn grows naturally in the cold desert without much human interference. Compared to many other fruit crops, systematic plantation and standard cultural practices are not followed. Therefore, seabuckthorn of Himalayan origin are 'wild harvests'

and do not contain traces of pesticides and other synthetic chemicals. After transferring a seabuckthorn-based technology by the Defence Institute of High Altitude Research (DIHAR) to a private firm in 2001, the collection of berries has become an important income-generating activity among local people. The collection period is short, and the return is high. Large-scale cultivation of seabuckthorn has the potential to be a key means for sustainable development of the cold desert of Ladakh. A systematic seabuckthorn plantation at a spacing of 3.0×1.0 m would accommodate 3,300 plants (female: 2,970; male: 330) per hectare. Assuming that each female plant bears 5.0 kg of berries yearly, 14.87 tons can be harvested annually per hectare. In comparison, seabuckthorn cultivars Jivko and Chuiskaya are reported to yield 19.0 and 25.1 tons per hectare, respectively, when planted at a spacing of 3.0×1.0 m in Russia (Khabarov, 2003). If cultivation is done on 2,500 hectares in a planned manner, the projected net income from seabuckthorn alone is estimated to be Rs 491 crore in the year 2030 (Bindal, 2016). Income generation will increase many-fold if value-added products are also manufactured in the region.

Income Generation

Seabuckthorn collection has been an important activity in the cold desert region of Ladakh since 2001. In 2013, 0.8% of the total population of the Leh district directly benefited from this berry collection. Each

individual collector collected 254.8 kg of berries on average, resulting in a net income of Rs. 8,154 per person within 5-10 days. However, those who worked for 12.6 days during the season earned an average of Rs. 18,375 per person (Stobdan et al., 2011b). In today's terms, that would equal Rs. 25,838 per person. Berries are sold at a premium price, with farmers receiving Rs 60 per kg in 2023. Many people who collect berries in Ladakh are from disadvantaged sections; most are women (67.4% of the workforce). Therefore, seabuckthorn has become an important source of sustainable livelihood in the region. Less than 10% of the total seabuckthorn resource in the area is being harvested. If the thick, dense seabuckthorn forests, which are currently inaccessible, were to be converted into productive stands with easy access for berry harvesting, it could greatly benefit the needy section of society. The Forest Department and local communities cultivate seabuckthorn on barren land for greening and income.

Cold desert is the major producer of seabuckthorn, which creates new opportunities for agri-entrepreneurs to process and add value at a commercial scale. Many unemployed, educated youth are looking for region-specific entrepreneurship opportunities. Commercial-scale development of value-added products can provide such income for them.

Ideal Plant for the Cold Desert

The ideal plant for the cold desert should have specific characteristics. It should be able to thrive in harsh conditions without much human interference and have ecological and economic benefits. Seabuckthorn is an ideal plant for large-scale plantations in the cold desert. The shrub has several advantages over other plants.

Seabuckthorn is native to the cold desert Ladakh region and thus well adapted to harsh conditions. The adventurous Moorcroft stayed for two years in Ladakh, from September 1820 to September 1822; he stated in his travelogue: *"The only wild fruit I observed was a small round orange-coloured berry, which was produced in great abundance by a prickly shrub called Chirma, or Chasta Ruru. The fruit is too acid to be eaten, although the flavour is agreeable, but the plant flourishes through the winter, and the berries are a great resource to the smaller birds"* (Moorcroft & Trebeck, 1837).

The shrub withstands extreme temperatures and drought, making it suitable for cultivation on vast uncultivated land in Ladakh's cold desert. Seabuckthorn has a symbiotic association with *Frankia*, which enables it to fix atmospheric nitrogen and improve soil fertility. The good news is that this plant is generally unaffected by major insect, pest, or

disease problems, so it requires relatively low management.

A major issue in the cold desert region is poor fruit yield due to insufficient pollinating insects during flowering. There are separate male and female plants in seabuckthorn, meaning pollination is necessary for fruiting. However, the flowers lack nectar and fragrance to attract pollinating insects. Interestingly, pollinating insects are not needed for pollination in seabuckthorn. The wind is the exclusive mode of pollination, with the pollen traveling up to 15 m in distance (Mangla & Tandon, 2014).

Seabuckthorn is a plant well-suited for sustainable development in cold desert regions. It thrives in harsh conditions and is both ecologically and economically important. If cultivated on 2,500 hectares in a planned in Ladakh, seabuckthorn alone is estimated to generate a projected net income of Rs 491 crore. Despite its potential, large-scale planting of this shrub has not been attempted in the cold desert.

Chapter-13

Seabuckthorn in India: Recent Developments

In India, seabuckthorn has received increasing attention after the Defence Institute of High Altitude Research (DIHAR), formerly Field Research Laboratory (FRL), developed and commercialized the technology for preparing beverages from its highly acidic fruit (Stobdan et al., 2009). DRDO has pioneered seabuckthorn research in India and started several R&D projects since the early nineties. Several research institutes have been working on the lesser-known shrub in recent years.

Several initiatives have been taken at the national and regional levels to promote the wonder plant. A study was commissioned by the Ministry of Agriculture and Farmers' Welfare, Government of India in 2017. Based on the study report, seabuckthorn has been included as a horticultural activity under the Mission for Integrated Development of Horticulture (MIDH) scheme in five Himalayan States/Union Territories. Leh district in the Union Territory of Ladakh and Lahaul & Spiti in Himachal Pradesh have launched schemes to boost the production and marketing of

seabuckthorn-based products. Seabuckthorn growing in Ladakh has received a GI tag in 2023. Considering the importance of seabuckthorn for the Himalayan states, a Center of Excellence (CoE) on Seabuckthorn is set up in Ladakh. Today, over hundred seabuckthorn-based foods and cosmetics products made in India are available online.



Center of Excellence (CoE) on Seabuckthorn in Ladakh

Once considered a thorny menace, seabuckthorn is now being seen as a means for sustainable development in the mountain region.

Mission for Integrated Development of Horticulture

A study titled 'Value Chain Analysis of Seabuckthorn in Leh Ladakh' was commissioned by the Ministry of Agriculture and Farmers' Welfare, Government of

India, in 2017. This study was conducted to explore the immense potential of seabuckthorn for the sustainable development of the Ladakh region. The study analyzed each link in the value chain of seabuckthorn in the region. It recommended ways to effectively use the existing resources and improve the marketability of seabuckthorn-based products. The study also undertook missing links that deter the cultivation of seabuckthorn. Based on the study report, seabuckthorn is now included as a horticultural activity under the Mission for Integrated Development of Horticulture (MIDH) scheme in Ladakh, Himachal, Uttarakhand, Arunachal, and Sikkim since April 2018. Farmers and processors are getting incentives from the Horticulture Department to produce planting material, establish orchards, create water resources, integrate post-harvest management and processing, and establish marketing infrastructure. There is also a research component where agencies can avail 100% funding for undertaking projects related to seabuckthorn.

One District One Product (ODOP)

ODOP initiative was initiated for regional development across all districts of the country. The initiative aims to select, brand, and promote at least one product from each district of the country to enable holistic socioeconomic growth across all regions. Under the ODOP initiative, Leh district in the Union Territory of Ladakh and Lahaul & Spiti in Himachal Pradesh have

selected seabuckthorn, intending to boost the production, marketing, and export potential of seabuckthorn-based products. The promotion and development of seabuckthorn-based products are expected to contribute to the districts' economic growth and preserve and promote India's rich cultural diversity. Under the scheme, the Leh district focuses on skill development, capacity building, infrastructure development, and marketing support for the identified product. These components aim to empower local farmers and small-scale industries by enhancing the quality and marketability of their products. The scheme provides financial assistance through loans and subsidies to eligible applicants, enabling them to grow their businesses and improve their product offerings.

GI-tag

A geographical indication (GI) is a sign used on products with a specific geographical origin and possess qualities or a reputation due to that origin. It identifies goods with unique characteristics originating from a definite geographical territory. Examples of Indian GI-tag include Basmati rice, Darjeeling tea, and Kanchipuram silk saree. GI-tag confers legal protection to the product and prevents unauthorized use of the registered product by others. Besides, it provides legal protection to the regional GI-tag products, which boosts exports. Most importantly,

it promotes economic prosperity for producers in a geographical territory.

Due to the region's climatic conditions, seabuckthorn growing in Ladakh has an exceptional quality or reputation. Thus, it received a GI tag on 22 June 2023. The GI tag grant has increased seabuckthorn products' visibility and promoted higher returns to the producers. Given this possibility, producers have incentives to improve their production. The Spiti Seabuckthorn Society, Himachal Pradesh, has also filed an application for GI tagging of Spiti seabuckthorn.

Center of Excellence

Center of Excellence (CoE) is a one-stop resource center that provides leadership, best practices, research, support, training of trainers, and skill training for a specific sector(s). It addresses emerging skill gaps and aligns training and research with industry needs. Considering the importance of seabuckthorn for the Himalayan states, a CoE on Seabuckthorn was set up under the MIDH scheme of the government of India. The CoE is located adjacent to Sangam, the confluence of the Indus and Zaskar rivers, approximately 35 km from Leh town.

Product Diversification

Leh Berry juice was the first seabuckthorn-based product launched in the Indian market in 2001. The

Ladakh-based DRDO-Defence Institute of High Altitude Research developed the product formulation. Today, over 100 seabuckthorn-based foods and cosmetics products made in India are available online. Food products include juice, powder, seed oil, pulp oil, jam, nutrition bar, tea, dried berries, and pulp. Bathing bar, body lotion, massage oil, after-bath oil, hair oil, shampoo, hair conditioner, face mask, face cream, face pack, face wash, face cleanser, face scrub, massage cream, skin nourishing cream, sunscreen, anti-wrinkle serum, and lipstick are some of the cosmetic products available commercially. Besides, health and hygiene products such as mouthwash and hand rub are also available in the market. Several industries and startups are currently manufacturing and marketing seabuckthorn-based products.

DUS Testing

Distinctness, Uniformity, and Stability (DUS) testing is a way of determining whether a newly bred variety differs from existing varieties within the same species (Distinctness), whether the characteristics used to establish distinctness are expressed uniformly (Uniformity) and whether these characteristics do not change over subsequent generations (Stability). DUS tests exist so that new varieties can legally gain access to their market and for the granting of Plant Breeders Rights, a form of intellectual property rights designed to safeguard the substantial economic investment involved in modern plant breeding.

A DUS test is usually conducted in the field over two successive growing seasons. During this period, several morphological characteristics are recorded both on the new (or candidate) variety and on similar varieties in what is known as 'Common Knowledge'. Differences, if they exist, are established by observation and measurement using agreed-upon protocols.

The DUS test guideline for seabuckthorn (*H. rhamnoides* L.) was developed by a Task Force constituted by the Protection of Plant Varieties and Farmers' Rights Authority, a statutory body created by an Act of Parliament. With the DUS guidelines in hand, any person can now apply for registration of new seabuckthorn varieties. The Office of the Registrar, Protection of Plant Varieties and Farmers' Rights Authority is appropriate for applying in India. Dr. YS Parmar University of Horticulture & Forestry, Solan, Himachal Pradesh, is the leading DUS testing center. DRDO-Defence Institute of High Altitude Research, Leh Ladakh, is the collaborating DUS test center.

Organic Certification

The Administrative of the Union Territory of Ladakh launched a Rs 500-crore organic farming project named 'Mission Organic Development Initiative of Ladakh' on 4 July 2020. The project also aims to realize the goals of a carbon-neutral Ladakh. It seeks to transform agriculture in Ladakh into a sustainable,

remunerative, respectable occupation and enable farmers to reap the benefits of dynamic market opportunities. Seabuckthorn in Ladakh is collected from its natural habitat. There are no herbicides, pesticides, or other chemicals, either natural or synthetic. Accordingly, seabuckthorn has been given priority crop status for organic certification in Ladakh.

Seabuckthorn gained prominence in 2001 with the launch of Leh Berry juice in the market. Since then, various initiatives have been taken at national and regional levels to promote the plant's benefits. Including the plant under the MIDH scheme and establishing a Center of Excellence on seabuckthorn is a significant boost to the industry. The recent grant of a GI tag to Ladakh's seabuckthorn has increased the visibility of the plant. The administrations in Ladakh and Lahaul & Spiti have launched a scheme to enhance the production and marketing of seabuckthorn-based products. With government support, the seabuckthorn industry is expanding in the country.

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About The Author



Dr. Tsering Stobdan, an agriculture scientist based in Ladakh, embarked on his educational journey at the Tibetan Children Village, Ladakh, where he received his formal education from pre-primary to secondary school. He then pursued his passion for horticulture, earning his Bachelor's degree from the

esteemed Dr YS Parmar University of Horticulture and Forestry, Solan. His thirst for knowledge led him to further his studies, culminating in a Master's degree from GB Pant University of Agricultural Science and Technology, Pantnagar, and a Ph.D. in Molecular Biology and Biotechnology from the prestigious Indian Agricultural Research Institute, New Delhi.

Dr. Stobdan's unwavering dedication to his work has placed him at the forefront of providing appropriate agro-technologies to the farmers in Ladakh. The technologies he developed have reached over 130 villages in Ladakh. Based on his report, Seabuckthorn is included as an activity under the Mission for Integrated Development of Horticulture (MIDH) scheme of GoI. He identified native apricots (Raktsey Karpo) as the world's unique and sweetest apricot, which later became Ladakh's first GI tag product. He introduced watermelon and sun melon as a new cash crop in Ladakh.

Dr. Stobdan's impactful research, which has led to the publication of over 90 research articles on topics related to horticulture and protected cultivation in Ladakh, is a testament to his significant contributions. Ten Research Fellows have been awarded Ph.D. degrees under his supervision.

Dr. Stobdan is a fellow of the Indian Academy of Horticultural Sciences (IAHS) and the Indian Society of Horticultural Research and Development. He is also an executive member of the Seabuckthorn Association of India. Dr. Stobdan is a recipient of the Fakhruddin Ali Ahmed Award for Outstanding Research in Tribal Farming Systems from the Indian Council of Agricultural Research. On 21 April 2022, the Ladakh Greenhouse technology that he developed received the Prime

*Minister Award for Excellence in Public Administration
(under the Innovation category).*